

**PUBLIC DEBT FORECASTING IN KENYA:
INTEGRATING MACROECONOMIC
VARIABLES WITH ARIMAX AND
EXTREME GRADIENT BOOSTING**

NDEGWA CATHERINE MURE

**A RESEARCH PROPOSAL SUBMITTED TO THE
FACULTY OF SCIENCE AND TECHNOLOGY IN PARTIAL
FULFILLMENT OF THE REQUIREMENT FOR THE AWARD OF
A
DEGREE OF MASTER OF SCIENCE IN STATISTICS,
MULTIMEDIA UNIVERSITY OF KENYA**

2025

DECLARATION

This proposal is my original work and has not been submitted for award of a degree in any other University.

Signature..... Date.....

NAME: Ndegwa Catherine Mure

This proposal has been submitted for examination with our approval as the University Supervisors.

Signature..... Date.....

Dr. Anthony Karanjah

Multimedia University of Kenya

Signature..... Date.....

Dr. Pius Kihara

Technical University of Kenya

DEDICATION

This work is lovingly dedicated to my father, John Ndegwa, my mother, Joyce Ndegwa, my brother, Moses Kirugu, and my sister, Charlene Wangari, whose unwavering love, constant support, and steadfast encouragement have been my greatest source of strength and inspiration throughout this journey. I am forever grateful.

ACKNOWLEDGEMENT

I am profoundly grateful to the Almighty God for His boundless grace, wisdom, and strength that have sustained me throughout the course of my Master's program. It is through His divine guidance that I have been able to persevere, grow, and accomplish this significant milestone in my academic journey. I extend my sincere appreciation to the Department of Mathematics at Multimedia University of Kenya for fostering a supportive environment. The resources, expertise, and encouragement provided by the faculty played a pivotal role in shaping my academic experience and facilitating the successful completion of this research. My deepest gratitude goes to my research supervisors for their invaluable guidance, thoughtful insights, and consistent support throughout this project. Their constructive feedback, patience, and commitment to academic excellence greatly enhanced the quality of my work and contributed significantly to my growth.

I am especially thankful to my father, John Ndegwa, and my sister, Charlene Wangari, for their generous financial support that made my studies possible. Your unwavering belief in my potential, your love, and your tireless encouragement uplifted me during moments of doubt and inspired me to press on even during the most difficult times. I would also like to extend heartfelt thanks to my friend, Frankline Mobagi, for his steadfast support and willingness to assist whenever I had questions regarding academic matters. His help and reassurance were a true source of relief and motivation during stressful times.

To all my family, friends, and peers who offered their encouragement, prayers, and moral support along the way, I am truly grateful. Each of you played a role in this achievement, and I carry this accomplishment not just as my own, but as a shared success.

ABSTRACT

In recent years, public debt in Kenya has been increasing rapidly, raising concerns about its sustainability and economic implications. As of July 2024, the public debt was roughly 10.6 trillion Ksh, reflecting a 2.9 percent increase from June 2023 when it stood at 10.3 trillion Ksh (Statista, 2024). This trend highlights the urgent need for accurate forecasting to ensure effective debt management and economic stability. This study aims to forecast Kenya's public debt by comparing a statistical model (ARIMAX) and a machine learning model (XGBoost). Macroeconomic indicators i.e. GDP, inflation, exchange rate and interest rate, are incorporated to analyze their impact on debt trends and evaluate the predictive performance of both ARIMAX and XGBoost. Additionally, scenario simulation is employed to assess how varying economic conditions affect public debt trajectories. The decision to compare ARIMAX and XGBoost is based on their distinct strengths and modeling assumptions. ARIMAX is effective in capturing linear relationships and temporal dependencies, while XGBoost is capable of detecting complex non-linear interactions and hidden patterns in the data. By comparing these two methods, the study aims to evaluate their respective forecasting performance, understand the influence of macroeconomic indicators on public debt, and determine which approach provides more accurate and policy-relevant predictions under various economic scenarios. The study utilizes historical data from 2001 to 2023, obtained from the Central Bank of Kenya (CBK) and Kenya National Bureau of Statistics (KNBS). By integrating statistical modeling, machine learning, and scenario simulation, this research aims to provide a comprehensive forecasting framework to support policymakers in making informed decisions for sustainable debt management.

Contents

Declaration	i
Dedication	ii
Acknowledgement	iii
Abstract	iv
1 Introduction	1
1.1 Background	1
1.1.1 Trends in Kenya’s Public Debt	2
1.2 Problem Statement	4
1.3 Objectives	5
1.3.1 General objective	5
1.3.2 Specific objectives	5
2 Literature Review	6
2.1 Introduction	6
2.2 Previous Studies	6
2.3 Research Gap	12
3 Methodology	13
3.1 Research Approach	13
3.2 Data Collection	13
3.3 Data Preprocessing	14
3.4 Model Selection and Implementation	15
3.4.1 ARIMA with Exogenous Variables (ARIMAX)	15
3.4.2 Extreme Gradient Boosting (XGBoost)	17
3.4.3 Variable Importance and Selection	20
3.4.4 Performance Evaluation	20
3.5 Scenario Simulation	21

4	Results and Discussion	22
4.1	Introduction	22
4.2	Stationarity Testing	22
4.3	ARIMAX Model Implementation	24
4.3.1	Residual Diagnostics	25
4.4	XGBoost Model Implementation	27
4.5	Influence of Macroeconomic Variables on Public Debt	28
4.6	Determining the Optimal Model for Forecasting Kenya's Public Debt	30
4.6.1	Comparative Evaluation of both ARIMAX and XGBoost Models	30
4.7	Future Predictions for XGBoost and ARIMAX Optimal Models	33
4.8	Scenario Simulation	34
4.9	Discussion	40
5	Conclusions and Recommendations	44
5.1	Conclusions	44
5.2	Recommendations	46
5.2.1	Policy Recommendations	46
5.2.2	Recommendations for Further Research	47

List of Tables

3.1	Data Sources and Descriptions	13
4.1	ARIMAX Model Coefficients	24
4.2	ARIMAX Model Coefficients for Exogenous Variables	29
4.3	Performance Metrics of Forecasting Models	31
4.4	Comparison of Actual and Predicted Yearly Total Public Debt (in KSH Millions)	32
4.5	Performance Metrics of Models for Univariate and Multivariate Data	32
4.6	Forecasted Public Debt in Kenya (2024–2028) using ARIMAX and XGBoost Models	34
4.7	Forecasted Total Public Debt under Different Scenarios	39
4.8	Forecasted Debt-to-GDP Ratio under Different Scenarios	39
4.9	Forecasted Debt-to-Interest Rate Ratio (ARIMAX) under Different Scenarios . .	39
4.10	Forecasted Debt-to-Exchange Rate Ratio (XGBoost) under Different Scenarios .	39

List of Figures

3.1	Conceptual Framework of XGBoost Model for Public Debt Forecasting	19
4.1	Differenced Total Debt over Time	23
4.2	Residuals vs Fitted Values	26
4.3	Histogram of Residuals	26
4.4	Autocorrelation of Residuals	27
4.5	Feature importance from XGBoost model	29
4.6	Actual Vs Predicted Public Debt (ARIMAX and XGBoost With Macroeconomic Variables)	30
4.7	Actual Vs Predicted Public Debt for both Univariate and Multivariate Models .	31
4.8	Future Predictions plot for XGBoost and ARIMAX Optimal models	33
4.9	ARIMAX Total Debt Scenarios	36
4.10	XGBoost Total Debt Scenarios	36
4.11	ARIMAX Debt-to-GDP Ratio	37
4.12	XGBoost Debt-to-GDP Ratio	37
4.13	ARIMAX Debt-to-Interest Rate Ratio	38
4.14	XGBoost Debt-to-Exchange Rate Ratio	38

Chapter 1

Introduction

1.1 Background

Public debt accumulation has become a major concern for many developing countries including Kenya. In Kenya, it has significantly increased over time due to the need to source affordable loans with fewer risks. These loans are mainly used to finance development projects, public services, and infrastructure Makau et al.,(2018). Still, they can pose serious risks to the economy's stability if not properly managed. Public debt arises when tax revenues and income from other sources are lower than what the government spends Robert J. Barro, (1979). Since the total income cannot compete with the ever-increasing government expenditure, this leads to public debt, both domestic and external. In Kenya, public debt is linked to macroeconomic factors such as GDP, inflation, exchange rate, and interest rate. GDP indicates the overall performance of the economy. It measures the monetary value of final goods and services produced in a given period hence determining whether the country can generate revenue to service debt. Inflation is the rate of increase in prices over a given period. It influences the cost of living which in turn can affect fiscal policies and debt sustainability. Exchange rate is the rate at which one currency can be converted into another. It dictates how much of one currency you need to get a certain amount of another. Interest rate, on the other hand, is the percentage cost of borrowing money, paid to the lender. Countries with low growth rates and high inflation struggle to sustain debt leading to increased borrowing costs, fiscal crisis, and currency devaluation (World Bank, 2021); (IMF, 2022). Depreciating exchange rate and rising interest rate increase the burden of debt making repayment more challenging (KIPPRA, 2021). Several studies have been employed to model and forecast public debt, with the most common methods being time series analysis and econometric models. ARIMA has been widely used due to its effectiveness in modeling temporal dependencies. Studies by Robert J. Barro, (1979) and F. Keraro et.al, (2024) have demonstrated

the use of this model in analyzing debt trends and making policy recommendations. ARIMAX extends ARIMA by including macroeconomic factors and this makes it more suitable for forecasting public debt. However, this method may fail to capture non-linear relationships between the variables, especially in the context of economic shocks such as the COVID-19 pandemic, recessions or structural changes. In recent years, machine learning techniques such as XGBoost and LSTM Networks have proven their ability to handle large datasets and non-linear relationships. These methods identify complex patterns and interactions in data that traditional statistical models may miss. For example, XGBoost has been used successfully for loan default prediction, offering high accuracy in identifying risky customers in financial data R. Odegua, (2020). Despite the advances in forecasting methodologies, the role of macroeconomic variables in shaping public debt trajectories has yet to be comprehensively analyzed F. Keraro et.al, (2024). This study focuses on addressing these gaps that can provide policymakers with robust tools for sustainable debt management in Kenya. By comparing these 2 models on extensive historical debt in Kenya, the study aims to identify the optimal model for forecasting the nation's future debt.

1.1.1 Trends in Kenya's Public Debt

Kenya has been on a rapid upward trajectory in regard to public debt over the last decade. The current debt-to-GDP ratio, 69.9%, has exceeded the recommended threshold of 50% or 55%, by the IMF for developing economies. Previous studies have suggested that a high debt burden could slow economic growth and increase vulnerability to external shocks. Kenya went on an expansionary fiscal policy which was driven by huge spending on infrastructural development. This led to an increase in gross public debt from 44.4 percent of GDP by end of 2010 to an estimated 79 percent of GDP by end of 2021. In 2020, Kenya's debt problem was mainly due to the COVID-19 global shock. The annual net borrowing since the 2013/14 financial year has increased by Ksh 685.72 Billion to reach Ksh 7.25 Trillion by November 2020. This indicates a 300 percent increase in debt stock between 2013 and 2020 (IEA Kenya).

Most of Kenya's external debts are on concessional terms but its commercial component has increased. By the end of 2019, external debt was about 31.5 percent of GDP evenly divided among multilateral, bilateral, and commercial creditors. Multilateral accounted for 33.5 percent, while bilateral accounted for 33 percent. Of this bilateral debt, about 72 percent is owed to non- Paris club members, mainly due to loans from China which financed the construction of Standard Gauge Railway(SGR). In October 2015, Kenya contracted a two-year USD 750 Million syndicated loan at LIBOR (London Interbank Offer Rate). Payment of almost 90 percent of this loan was extended to 7 years. In January 2019, Kenya contracted another 10 year USD

250 Million syndicated loan and a 9 year USD 1.25 Billion loan for refinancing purposes (World Bank).

Domestic debt is issued mostly in form of treasury bonds and bills. Increased domestic debt has been necessitated by a widening fiscal deficit. Domestic debts formed a large share of Kenya's debts between the financial years of 2012/13 and 2013/14. However, since 2016/17 financial year, the government started depending majorly on international markets for fundings leading to a decrease in domestic stock compared to external. Domestic debt rose from Ksh 167 Billion in 2012/13 to Ksh 251 Billion in 2014/15, fiscal deficit had increased from 5.1 percent to 8.4 percent respectively. This debt is compared of pre-1997 government debt, CBK overdraft, tax reserve certificates and commercial bank. CBK held 4.3 percent of the total domestic debt between 2012/13 and 2017/18 while commercial banks and non-bank financial institutions accounted 50.9 and 44.8 percent respectively (Kippra, 2019). At the end of 2019, Kenya's domestic public debt was 30.2 percent of GDP which is an increase from 27.4 percent in 2010 (World Bank).

1.2 Problem Statement

Public debt levels in Kenya have been rising at an alarming rate, raising concerns about economic sustainability and fiscal stability. While borrowing is essential for financing development projects and stimulating growth, excessive debt accumulation can lead to economic distress, currency depreciation, and reduced investor confidence. In Kenya, the increasing debt burden has resulted in over-taxation and rising costs of essential commodities, ultimately lowering living standards and slowing economic growth Kithure, (2021). Recent data shows that Kenya's debt-to-GDP ratio has exceeded the recommended threshold of 34.7 percent, reaching 69.87 percent as of 2024 (Statista, 2024). This sharp rise has raised serious concerns over debt sustainability, signaling the urgent need for accurate forecasting models to guide proactive debt management strategies. Traditional statistical forecasting models, such as ARIMA and VAR, have been widely used to predict public debt trends. However, these methods struggle to incorporate macroeconomic factors like the GDP, inflation, exchange, and interest rates, which are key drivers of debt sustainability. Additionally, these models assume linear relationships and stable economic conditions, making them less effective in capturing sudden economic shocks or policy changes. Machine learning techniques, despite their success in other financial forecasting applications, remain largely underexplored in Kenya's public debt forecasting. These models offer superior accuracy in capturing non-linear trends and complex interactions between macroeconomic variables, yet their potential has not been fully harnessed. Furthermore, while scenario simulation is widely recognized as a critical tool for debt risk assessment, it is rarely integrated into forecasting models. Without scenario-based forecasting, policymakers lack insights into how different economic conditions (e.g., inflation surges, GDP fluctuations, depreciating exchange rate) impact debt sustainability. This gap limits the effectiveness of existing debt management strategies, leaving the economy vulnerable to financial shocks. This study seeks to address these challenges by using ARIMAX and XGBoost models in forecasting Kenya's public debt, while integrating macroeconomic variables as predictors. Additionally, scenario simulation will be employed to assess the impact of varying economic conditions on debt sustainability, providing a more comprehensive and adaptable forecasting framework. By bridging these gaps, this study aims to provide data-driven insights for fiscal policy, debt sustainability, and economic planning in Kenya.

1.3 Objectives

1.3.1 General objective

To forecast public debt in Kenya by integrating macroeconomic variables using ARIMAX and Extreme Gradient Boosting methods.

1.3.2 Specific objectives

- I. To implement ARIMAX and XGBoost debt prediction models that incorporate the macroeconomic variables (GDP, Inflation, exchange and Interest rates).
- II. To evaluate the performances of both models in forecasting and identify the most influential macroeconomic variables affecting public debt.
- III. To apply scenario simulation to evaluate the impact of varying macroeconomic variables on public debt forecasts.

Chapter 2

Literature Review

2.1 Introduction

Forecasting of public debt in Kenya is critical for maintaining fiscal sustainability and economic stability. Accurate predictions enable governments to prepare effectively for debt-related crisis and design interventions to ensure sustainability. This section presents previous research on public debt forecasting, particularly focusing on Kenya and their relevance to this study. It also addresses the methodological gaps addressed in this study.

2.2 Previous Studies

Various studies have used different methodologies to predict public debt trajectories, each with its strengths and limitations.

Ahmed et al. (2020) introduced a hybrid time series forecasting model that combined ARIMA and XGBoost to capture both linear and non-linear patterns in economic data. The study found that this hybrid model outperformed both individual models across several forecasting tasks by balancing trend estimation and complex pattern recognition. In the context of public debt, such a hybrid approach could be valuable, as both gradual fiscal trends and abrupt macroeconomic events often drive debt accumulation. The success of the ARIMA-XGBoost hybrid supports the exploration of machine learning-enhanced forecasting in government finance.

ARIMA models have been used widely for time series forecasting. F. Zhuravka et.al (2019) used this model in a study focusing on Ukraine's government debt to predict future debt values. Results showed that the model was effective in capturing trends and cyclic movements in debt accumulation making it reliable for short term forecasting. However, it was noted that the model

couldn't effectively predict sudden economic shocks.

Mutuku and Koech, (2021) conducted a comprehensive study on forecasting Kenya's public debt using both the ARIMA and Vector Autoregressive (VAR) models. Their research demonstrated that while ARIMA was effective in modeling univariate debt series, VAR offered a more robust framework by incorporating multiple macroeconomic variables such as GDP growth, inflation, and interest rates. The study found that VAR models outperformed ARIMA in long-term forecasting due to their ability to reflect dynamic feedback among variables. This supports the rationale for using models like ARIMAX or multivariate ML approaches, which can accommodate external macroeconomic factors and better capture economic complexity in public debt trends.

Chen and Guestrin, (2016) carried out a study that found XGBoost to be 10 times faster than traditional boosting methods while achieving superior predictive performance. It consistently outperformed other ML models in classification and regression tasks. Due to its ability to handle non-linear relationships it is a strong candidate for debt forecasting especially when incorporating external factors like GDP and Inflation.

In another study, Mbiti and Njoroge, (2020) explored the influence of key macroeconomic indicators, particularly inflation, fiscal deficits, and exchange rate volatility, on the dynamics of public debt in Kenya. Although their work did not focus on forecasting, their findings are critical in identifying relevant variables that impact public debt. These macroeconomic drivers are useful for enhancing forecasting accuracy, particularly in models like ARIMAX and XGBoost that allow the integration of exogenous predictors. The study underlines the need for more sophisticated forecasting models that reflect the influence of fiscal and monetary indicators on debt evolution.

Keraro et.al (2024) did a study comparing ARIMA and Holt-Winters model to forecast Kenya's public debt, both models provided accurate short-term forecasts with Holt-Winters performing better when seasonality was present in the data. However, ARIMA was more effective in long-term trend estimation. From findings, the model assumed that past patterns would continue, making it ineffective for handling irregular economic shocks.

Makridakis et al. (2018) compared statistical and machine learning forecasting approaches across multiple domains. They concluded that machine learning models, including gradient boosting methods like XGBoost, often outperform traditional models in large, noisy, and non-linear datasets. However, they also pointed out that classical models such as ARIMA remain valuable due to their transparency and ease of interpretation, qualities that are especially important in public sector decision making. This underscores the potential for combining these models or

applying them in parallel to benefit from both accuracy and interpretability.

Nichani et.al (2024) did a research focused on optimizing financial time series predictions by combining ARIMA with LSTM and XGBoost models. All 3 models produced good results but the ARIMA-LSTM hybrid outperformed individual models. It achieved on MSE of 22.15 and an R-squared value of 0.99. These results highlight the potential of integrating traditional and ML techniques for enhanced prediction accuracy.

The interpretability of machine learning models has been a growing concern, especially in policy-making environments. Doshi-Velez and Kim, (2017) emphasized the importance of explainability in machine learning applications, particularly when used in sensitive public domains like finance. Their work argues for a balance between predictive performance and model transparency, supporting the use of ARIMA or ARIMAX for their clarity while leveraging the predictive power of models like XGBoost where high accuracy is critical.

Techniques such as XGBoost and LSTM have been applied widely in financial forecasting but there is limited research on their application in public debt forecasting. However, Daniel, (2018) did a study on U.S treasury interest rate forecasting that demonstrated their effectiveness. From the study XGBoost handled complex, non-linear relationships therefore outperformed traditional econometric models in predicting interest rate. LSTM networks on the other hand were found to be effective for long term sequence prediction.

Uhumwangho, (2024) did a study that investigates the effectiveness of XGBoost and LSTM models in predicting stock price movements. The results showed that XGBoost achieved a Directional Accuracy Percentage (DAP) of 71.02% while LSTM had a DAP ranging from 62.42% to 67.10%. This shows that XGBoost was more effective, further demonstrating its predictive strength. These findings suggest that XGBoost could offer similar advantages in public debt forecasting, especially when macroeconomic volatility is a factor.

Wambugu, (2022) applied ARIMAX models to forecast inflation in Kenya and found that including external factors such as oil prices and exchange rates improved model performance compared to traditional ARIMA. Her findings are especially relevant to this study, as they confirm that ARIMAX models offer greater flexibility and accuracy in forecasting economic indicators influenced by external shocks. The success of ARIMAX in macroeconomic forecasting reinforces its applicability to public debt modeling, particularly in contexts where debt is sensitive to global and domestic economic fluctuations.

A methodological contribution by Agung, (2009) further emphasized the importance of incorpo-

rating exogenous variables in time series forecasting. His work provides an applied framework for using ARIMAX models in economic policy analysis, particularly in emerging economies. Agung's approach highlights how macroeconomic indicators can be integrated into forecasting models to enhance their responsiveness to policy shifts and global market trends. This is especially relevant for Kenya, where debt levels are often influenced by international borrowing, donor flows, and exchange rate variations.

Zulu and Mwansa, (2020) did a study comparing ARIMA, Moving Average(MA), and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models to focus Zambia's external debt. From the study, ARIMA produced the lowest error rates (MAPE and MSE) making it the most reliable model. MA and GARCH were less accurate due to their inability to capture long term trends. However, ARIMA's performance declined when sudden macroeconomic shifts such as GDP and Inflation occurred.

Maina et al., (2024) used a Vector Autoregressive (VAR) framework to analyze interactions between inflation and public debt in Kenya. Although not ARIMAX specific, their results demonstrated significant dynamic feedback between macroeconomic indicators and public debt levels. This reinforces the need to use ARIMAX, which can also model exogenous effects within a time series structure.

Ugoh et al., (2021) applied an ARIMAX model to forecast Nigeria's GDP by incorporating imports as an exogenous variable. Their findings revealed that ARIMAX outperformed ARIMA, confirming that incorporating macroeconomic drivers significantly enhance predictive accuracy. This study supports the adoption of ARIMAX in settings where external shocks and fiscal drivers shape macroeconomic outcomes.

Njenga, (2024) focused on forecasting domestic credit growth using a seasonal ARIMA(2,1,2)(1,0,0)[12] model. The study found that ARIMA could effectively capture both trend and seasonal cycles. The forecast projected a steady growth rate in domestic credit into 2024, validating ARIMA's role in stable financial environments. This highlights ARIMA's utility but also suggests limitations in handling structural shifts or non linear macroeconomic influences—areas where models like XGBoost may provide an edge.

A study by Badrawani, (2025) demonstrated that XGBoost significantly outperformed ARIMA and SARIMA, in predicting inflation in Indonesia. The model reduced RMSE by approximately 45%, highlighting its strength in capturing non linear dependencies and structural shifts in macroeconomic data. This supports the role of XGBoost in macroeconomic forecasting where sudden changes in policy or global markets may disrupt historical trends.

Kashif and Ślepaczuk, (2024) implemented a hybrid Long Short Term Memory (LSTM)-ARIMA approach to forecast equity indices and optimize algorithmic trading strategies. Similarly, Shi et al., (2022) applied an attention enhanced CNN-LSTM-XGBoost model for stock price prediction, noting strong performance in volatile environments. Their findings validate XGBoost’s value when used as a comparative benchmark due to its reliable predictive performance across multiple metrics.

Although this study does not use hybrid models, a study by Omondi et al., (2024) supports the comparative analysis of ARIMAX and XGBoost by showcasing each model’s individual strengths. Their study combined ARIMA and XGBoost to forecast M-PESA transaction volumes in Kenya. From the study, ARIMA captured the trend while XGBoost modeled residual non linearities. This supports separate analysis of each model to understand what each contributes.

Kavousi-Fard et al., (2018) used an ARIMA-XGBoost hybrid to forecast China’s energy indicators. Although the hybrid showed superior performance, ARIMA contributed baseline forecasting accuracy while XGBoost enhanced responsiveness to volatility. Zhu et al., (2023) employed a wavelet-decomposition based ARIMA-XGBoost hybrid for commodity price and exchange rate forecasting, again validating the individual strengths of each model.

Owaga, (2021) did an empirical investigation which employed time series regression to examine the impact of budget deficits, trade balances, official development assistance (ODA), and GDP growth on Kenya’s public debt. The study found that budget deficits had a strong positive correlation with debt levels, while GDP growth had an insignificant effect. This reinforces the need to include such variables as exogenous inputs in debt forecasting models and supports the use of ARIMAX or XGBoost to capture their influence.

In another study by Cheruiyot et al., (2024), they applied an ARIMA(2,2,2) model to forecast inflation trends in Kenya using historical CPI data. The model achieved low root mean square error (RMSE) and mean absolute error (MAE), confirming its effectiveness in modeling macroeconomic variables with strong autocorrelation structures. Nonetheless, the study acknowledged that ARIMA may underperform during periods of economic instability or sudden fiscal shocks, emphasizing the importance of incorporating exogenous variables, which ARIMAX can accommodate.

Bentéjac et al., (2019) compared the performance of XGBoost with other ensemble learners such as Random Forest and traditional Gradient Boosting Machines (GBM). The study evaluated over 100 datasets and found that XGBoost generally achieved higher accuracy. The authors emphasized XGBoost’s robustness in handling missing data, multicollinearity, and noisy features,

making it ideal for economic forecasting tasks where data imperfections are common.

A study conducted by Masini et al., (2020) reviewed machine learning approaches in time series forecasting, with a particular focus on financial and economic data. The authors highlighted the strengths of tree based models like XGBoost in capturing non linear dependencies and interactions. They also noted that machine learning models, when properly tuned, can outperform traditional statistical methods in terms of accuracy and adaptability, especially when dealing with volatile macroeconomic variables.

Siami-Namini & Siami Namin, (2018) conducted a study of direct comparison between ARIMA and Long Short-Term Memory (LSTM) models on financial time series datasets. The LSTM model outperformed ARIMA with an 84% to 87% reduction in forecasting errors. The study concluded that machine learning models like LSTM, and by extension, XGBoost, are better suited for capturing long term dependencies and complex, non linear dynamics. Their findings support this study's motivation to explore machine learning as an alternative to classical statistical models in forecasting public debt.

Mukono & Mithamo, (2019) did a study on yield curve forecasting that applied the Nelson Siegel model to forecast Kenya's government bond yield curve, incorporating exogenous factors such as global interest rates and inflation. The study concluded that including such external variables significantly improved forecast accuracy. Although not a time series study, it supports the idea that macroeconomic indicators influence government debt instruments and should be considered in models like ARIMAX and XGBoost for better debt prediction accuracy.

Using a Bayesian Vector Autoregressive (BVAR) model, Caspah Lidiema et al., (2018) did a study that forecasted household credit in Kenya and compared its performance against univariate models. The BVAR approach, which allows the incorporation of prior distributions and multiple variables, outperformed ARIMA in forecast accuracy. The study underscores the value of multivariate forecasting frameworks in economic analysis.

2.3 Research Gap

Despite advancements in statistical and machine learning techniques for public debt forecasting, several gaps remain in the existing literature. While ARIMA-based models have been widely used, their inability to capture non-linear relationships and economic shocks limits their accuracy, especially in dynamic economic environments. On the other hand, machine learning models such as XGBoost and LSTMs, when combined with macroeconomic indicators, have demonstrated strong predictive capabilities in financial forecasting. However, their application in public debt forecasting remains largely absent, despite their potential to capture complex interactions between macroeconomic variables and debt trends. However, these models lack interpretability and may overfit in scenarios with limited historical data. The trade-off between accuracy and explainability remains an open challenge in debt forecasting. Additionally, scenario simulation is largely missing in public debt forecasting studies. Most models assume stable economic conditions, overlooking the effects of policy shifts, global economic crises, and domestic structural changes. Studies fail to account for varying GDP growth, exchange rate and interest rate, or inflationary pressures, yet these factors significantly impact debt sustainability. Integrating scenario simulation into forecasting models is essential for policymakers to anticipate risks and develop proactive debt management strategies. Furthermore, while GDP, exchange rate and interest rate, and inflation are widely recognized as key macroeconomic indicators, they have not been fully integrated into forecasting models. Many models treat these variables as external inputs rather than incorporating them as dynamic predictors that shape debt trajectories. A more structured approach that embeds these indicators within both statistical and machine learning models could enhance predictive accuracy and offer deeper insights into debt sustainability under changing economic conditions. This study addresses these gaps by implementing a forecasting framework that integrates these macroeconomic variables while incorporating scenario simulation to evaluate the impact of different economic conditions on debt sustainability. By bridging these gaps, the research provides a more comprehensive and adaptive approach to public debt forecasting in Kenya, equipping policymakers with reliable and scenario-responsive forecasting tools for effective debt management.

Chapter 3

Methodology

3.1 Research Approach

This study outlined the methodological framework used to forecast Kenya’s public debt. A comparative and predictive research approach integrating statistical (ARIMAX) and Machine Learning (XGBoost) techniques were employed to forecast Kenya’s public debt while incorporating GDP, Inflation, Exchange and Interest rates as the key macroeconomic indicators. Scenario simulation was also incorporated to assess the impact of varying economic conditions on debt sustainability.

3.2 Data Collection

This study relied on secondary data obtained from the official Central Bank of Kenya (CBK) and Kenya National Bureau of Statistics (KNBS) websites.

Table 3.1: Data Sources and Descriptions

Variable	Source	Description
Public Debt	CBK	Monthly and Annual figures for both Domestic and External Debt
GDP	CBK, KNBS	Annual GDP rates and Nominal GDP values
Inflation Rate	CBK, KNBS	Monthly Inflation rates to capture short-term economic fluctuations
Exchange Rate	CBK	Monthly and Annual figures
Interest Rate	CBK	Monthly and Annual figures

This study covered a 22-year historical dataset, i.e., the period 2001 to 2023, to capture long-term public debt trends and their relationship with macroeconomic indicators. This period ensured sufficient data for training predictive models while accounting for major economic events such as the 2008 global financial crisis and the 2020 COVID-19 pandemic.

3.3 Data Preprocessing

The following steps were applied: missing data was imputed, which ensured that missing values did not distort forecasts. Linear Interpolation method was used for public debt data to maintain trend consistency.

$$x_t = x_{t-1} + \frac{(x_{t+1} - x_{t-1})}{2} \quad (3.1)$$

where x_t is the missing value, estimated as the average of previous and next values while nearest neighbor imputation was used for the macroeconomic variable. Outliers that were present due to sudden borrowing were analyzed to verify their accuracy where extreme fluctuations in the variables that exceeded 1.5 times the interquartile range (IQR) from the quartiles were smoothed using simple moving average, to maintain data integrity while preserving trends. The formula for the simple moving average is;

$$S_t = \frac{1}{N} \sum_{i=t-N+1}^t X_i \quad (3.2)$$

where S_t is the smoothed value at time t , N , the number of periods used for smoothing, and X_i , the actual value at time i . Feature engineering was done to create lag features that helped enhance accuracy and capture historical dependencies.

$$\text{lag}_n = x_{t-n} \quad (3.3)$$

where lag_n is the value of the variable x from n periods ago. Stationarity test was carried out using the Augmented Dickey-Fuller (ADF) test to test for stationarity on the target variable, Public Debt. The ADF test is a widely used unit root test that evaluates the null hypothesis that a time series possesses a unit root (i.e., it is non-stationary), against the alternative hypothesis that the series is stationary. A p value of more than 0.05 implies non-stationarity. The basic autoregressive (AR) representation of order 1 for a time series y_t :

$$y_t = \alpha + \beta t + \rho y_{t-1} + \varepsilon_t \quad (3.4)$$

where α is the constant term, βt represents the deterministic time trend, ρ is the autoregressive coefficient, y_{t-1} is the lagged value of the series, and ε_t is the white noise error term. To derive the ADF regression form, we subtract y_{t-1} from both sides of equation (3.4):

$$\Delta y_t = y_t - y_{t-1} \quad (3.5)$$

$$= \alpha + \beta t + \rho y_{t-1} - y_{t-1} + \varepsilon_t \quad (3.6)$$

$$= \alpha + \beta t + (\rho - 1)y_{t-1} + \varepsilon_t \quad (3.7)$$

Letting $\gamma = \rho - 1$, we obtain the basic Dickey-Fuller (DF) regression:

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \varepsilon_t \quad (3.8)$$

To account for higher-order autocorrelation in the residuals, the Augmented Dickey-Fuller (ADF) test included lagged differences of the dependent variable. The resulting ADF regression model is:

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^p \delta_i \Delta y_{t-i} + \varepsilon_t \quad (3.9)$$

where Δy_t is the first difference of the time series y_t , γ is the coefficient of interest for testing stationarity, $\sum_{i=1}^p \delta_i \Delta y_{t-i}$ represents the lagged difference terms to correct for autocorrelation, and ε_t is a white noise error term.

The null hypothesis $H_0 : \gamma = 0$ suggests the presence of a unit root (non-stationarity), while the alternative hypothesis $H_1 : \gamma < 0$ indicates stationarity of the series.

3.4 Model Selection and Implementation

With the pre-processed data, both ARIMAX and XGBoost models were trained and evaluated to determine the optimal forecasting approach for Kenya's public debt. The selection of the best ARIMAX configuration was guided by Akaike Information Criterion (AIC), which balances model complexity and goodness of fit. Hyperparameter tuning was performed for XGBoost using grid search cross validation to optimize parameters such as learning rate, tree depth, and the number of estimators. The final model selection was based on performance metrics, including Root Mean Squared Error (*RMSE*), Mean Absolute Percentage Error (*MAPE*), and R-Squared (R^2) ensuring that the most accurate and reliable model is used for forecasting.

3.4.1 ARIMA with Exogenous Variables (ARIMAX)

ARIMAX model extends the ARIMA framework by incorporating exogenous variables, to improve forecasting accuracy. It is a multiple Linear Regression Model denoted by the ARIMAX(p,

d, q) i.e Autoregressive (AR) term [p] which captures past dependence in time series, Differencing [d] which ensures stationarity through elimination of trends in the data and Moving Average (MA) [q], which models dependencies on past forecast errors.

The general form of the ARIMAX model incorporating autoregressive (AR), integrated (I), and moving average (MA) components along with exogenous variables is given as:

$$\Delta^d y_t = c + \sum_{i=1}^p \phi_i \Delta^d y_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \sum_{k=1}^m \beta_k x_{k,t} + \varepsilon_t \quad (3.10)$$

where $\Delta^d y_t$ is the d -th order differenced target variable, c is the constant (intercept which represents the mean of the differenced series that is not explained by the autoregressive, moving average, or exogenous terms) term, p is the number of autoregressive lags with coefficients ϕ_i , q is the number of moving average terms with coefficients θ_j , $x_{k,t}$ represents the k^{th} exogenous variable at time t , m is the number of exogenous variables, β_k are the coefficients for the exogenous variables, ε_t is the white noise error term.

A first-order differenced public debt series can be written as:

$$\Delta PD_t = PD_t - PD_{t-1} \quad (3.11)$$

where PD_t is public debt at time t . The variables represent the historical values of public debt while time t is crucial for forecasting. ARIMAX relies on the following assumptions:

- I. The time series must be stationary, meaning mean and variance remain constant over time.
- II. The model assumes a linear relationship between the dependent variable and its past values.
- III. Independence of residuals i.e. the error terms should not exhibit autocorrelation.

To determine the optimal ARIMAX model, a grid search procedure was employed to explore various combinations of the non-seasonal model parameters (p, d, q) , where p is the autoregressive order, d is the order of differencing and q is the moving average order. The value of d was determined based on stationarity tests, particularly the Augmented Dickey-Fuller (ADF) test. Once differencing was applied to achieve stationarity, combinations of p and q ranging from 0 to 3 were systematically tested. This range was chosen to balance model simplicity and flexibility. Testing beyond this range may lead to overfitting without significantly improving forecast accuracy. For each candidate model, an ARIMAX model was fit using the macroeconomic variables (GDP, inflation, exchange rate, and interest rate) as exogenous regressors. The Akaike Information Criterion (AIC) was computed for each model.

The AIC is a widely used model selection metric that balances model fit and complexity. It provides a means of comparing different statistical models by evaluating how well they explain the data while penalizing models that are overly complex. AIC is defined as:

$$AIC = 2k - 2 \ln(L) \quad (3.12)$$

where k is the number of estimated parameters, L is the maximum likelihood of the model. Several models were evaluated, and the one with the lowest AIC value was chosen indicating a model that fits the data well. After fitting the ARIMAX model, residual diagnostics was performed to evaluate the adequacy of the model and verify that the underlying assumptions hold. Specifically, residuals were analyzed to assess whether they are randomly distributed, uncorrelated, and homoscedastic (i.e., have constant variance). These diagnostics involved visual and statistical techniques, including Residual vs fitted value plots to check for linearity, Histogram to assess the normality of residuals, Autocorrelation Function (ACF) plots to detect serial correlation, and Ljung-Box test to statistically assess independence of residuals.

The ARIMAX model provided coefficients for each of the 4 macroeconomic variable. The magnitude and statistical significance (p-values) of these coefficients indicated the strength and direction of each variable's influence on public debt.

3.4.2 Extreme Gradient Boosting (XGBoost)

XGBoost is a supervised learning algorithm based on gradient boosting trees and per research, it has shown exceptional performance in predictive tasks as shown by Chen and Qureshi, (2016). This machine learning technique is known for its ensemble learning approach and capability to handle structured data. XGBoost operates within the gradient boosting framework which focuses on optimizing speed and model performance. XGBoost incorporates L1 (Lasso) and L2 (Ridge) regularization techniques which prevent overfitting tendencies according to Uhumwangho, (2024). This technique is widely used for financial and economic forecasting due to its ability to capture complex, non-linear relationships and handle missing data efficiently. XGBoost builds multiple decision trees, with each correcting errors from the previous one. It is defined as:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i), \quad f_k \in F \quad (3.13)$$

where $\hat{y}_i$ is the predicted public debt at time i , $f_k(x_i)$ represents each decision tree, k is the number of boosting iterations, f is the set of regression trees, F is the space of regression trees used for boosting. Each tree function $f_k(x_i)$ can be expressed as:

$$f_k(x_i) = w_{q(x_i)} \quad (3.14)$$

where $q(x_i)$ maps the input x_i to the index of a leaf node in the tree, $w_{q(x_i)}$ is the score or weight assigned to that leaf.

The objective function minimizes the loss function while applying regularization

$$obj_j = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (3.15)$$

where $l(y_i, \hat{y}_i)$ is the loss function measuring prediction error, $\Omega(f_k)$ is the regularization term that prevents overfitting. For regression problems such as public debt forecasting, the loss function is commonly defined as the squared error:

$$l(y_i, \hat{y}_i) = (y_i - \hat{y}_i)^2 \quad (3.16)$$

The regularization function $\Omega(f_k)$ used to penalize model complexity is given by:

$$\Omega(f_k) = \frac{1}{2} \lambda \sum_{j=1}^{T_k} w_{kj}^2 + \alpha \sum_{j=1}^{T_k} |w_{kj}| \quad (3.17)$$

where T_k is the number of leaves in tree k , w_{kj} is the score on leaf j of tree k , λ and α are the L2 (Ridge) and L1 (Lasso) regularization parameters, respectively.

XGBoost requires careful selection of hyperparameters to optimize performance and prevent overfitting. Grid SearchCV was used to systematically test different combinations of hyperparameters and select the best-performing set based on evaluation metrics such as RMSE, MAPE, and R^2 . The key hyperparameters for tuning included learning rate (η) which controls how much the model adjusts with each iteration, maximum tree depth (d) which limits how complex each tree can be, number of estimators (n) which defines the number of boosting iterations, and $L1(\lambda_1)$ and $L2(\lambda_2)$ regularization which penalize overly complex models to prevent overfitting.

The inclusion of XGBoost in this study was motivated by the potential non-linearity in the relationship between public debt and macroeconomic indicators. In economic data, relationships are often not linear. The effect of different macro economic factors on public debt may vary depending on the rate of economic growth, external shocks, or fiscal responses. Non-linearity is observed when scatter plots between variables do not form a straight-line trend, or when ARIMAX residuals exhibit patterns rather than random dispersion. The use of both models in this study not only enabled a performance comparison but also helped assess whether non-linearity plays a significant role in Kenya's public debt dynamics. The XGBoost model was

analyzed using feature importance scores — a measure of how often and how strongly each variable contributes to tree-based splits during prediction. This allows for ranking variables based on predictive power, offering insight into which macroeconomic factors are most influential.

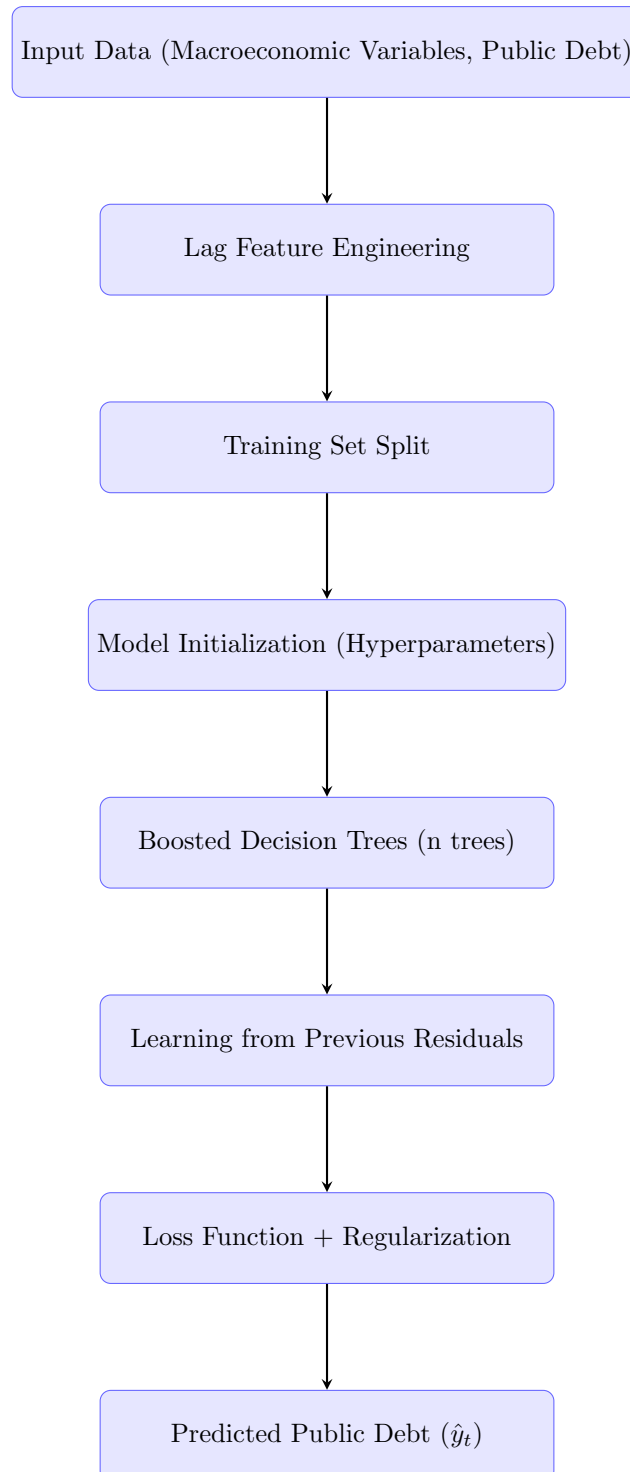


Figure 3.1: Conceptual Framework of XGBoost Model for Public Debt Forecasting

3.4.3 Variable Importance and Selection

For the ARIMAX model, importance was evaluated based on the statistical significance (p-values) and magnitude of the model coefficients. Variables with low p-values (less than 0.05) and larger coefficients were considered more influential in explaining changes in public debt over time. For the XGBoost model, variable importance was assessed using feature importance metrics, which measured the contribution of each variable to the predictive accuracy of the model. Variables that showed consistently high importance across both models were interpreted as having strong economic influence on public debt. The results of this analysis lead to policy recommendations by identifying which macroeconomic indicators should be prioritized in debt management strategies.

3.4.4 Performance Evaluation

This section shows how well the models perform. RMSE, MAPE and R^2 were the evaluation metrics used to assess the accuracy and effectiveness of ARIMAX and XGBoost.

Root Mean Squared Error (RMSE) was used to measure the average prediction error. A lower RMSE for XGBoost model indicated better forecasting performance. It is calculated using the formula:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (3.18)$$

where n is the number of observations / data points, y_i is the actual public debt value at time i , $\hat{y}_i$ is the predicted public debt value at time i . **Mean Absolute Percentage Error (MAPE)** expresses forecast error as a percentage of actual values and is preferred for comparing models across different scales. MAPE is robust to outliers since error margins are given as percentages rather than absolute deviations. A lower MAPE value for XGBoost indicated better accuracy. It's calculated by the formula:-

$$MAPE = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100 \quad (3.19)$$

where n is the number of observations, y_i is the actual public debt at time i , $\hat{y}_i$ is the predicted public debt value at time i . **R -squared (R^2) score** measured how well the models explained variability in public debt. Higher R^2 value of XGBoost indicated its stronger predictive power.

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2} \quad (3.20)$$

If $R^2 = 1$, it implies a perfect fit (i.e., the model explains all variation) and $R^2 = 0$ implies the model explains nothing beyond the mean debt value.

The model with the lowest RMSE and MAPE, and the highest R^2 was considered the most accurate forecasting method.

3.5 Scenario Simulation

This section features the methodology for assessing the impact of different economic conditions on public debt forecasts through scenario simulation. This is a crucial technique in forecasting because it allows policymakers to test different economic conditions and anticipate how public debt will evolve under varying macroeconomic scenarios. It helps answer questions such as: What happens to public debt if GDP accelerates? How does rising inflation impact debt sustainability? What is the impact of fluctuating interest or exchange rates on borrowing needs? This study evaluated 3 different economic scenarios. The first is **Baseline scenario** which assumes that GDP, inflation, interest rates, and exchange rates follow historical trends with no major disruptions. This scenario serves as the reference point for comparing other projections. The second is **Optimistic Scenario** which assumes accelerated GDP growth, a decrease or stabilization in inflation, and constant interest and exchange rates. The expected outcome is a stable or improving debt-to-GDP ratio and reduced risk of debt distress. The third is **Pessimistic Scenario** which assumes a slowdown in GDP growth and a significant increase in inflation, with interest and exchange rates held constant. The expected result is a higher debt burden, increased risk of default, and reduced fiscal space.

In this study, public debt forecasting was conducted using both ARIMAX and XGBoost models. These models were first trained on historical data under observed economic conditions. To simulate future scenarios, values of the variables were adjusted based on defined optimistic and pessimistic projections. The adjusted datasets were fed into both models to generate new forecasts under each scenario. The outputs were compared using key indicators such as the debt-to-GDP ratio, debt-to-interest rate and debt-to-exchange rate. Insights gained from these forecasts help inform fiscal policy, identify potential vulnerabilities and supporting the development of robust debt management strategies under various economic conditions.

Chapter 4

Results and Discussion

4.1 Introduction

This chapter presents the results of both ARIMAX and XGBoost models developed to forecast Public Debt in Kenya using GDP, Inflation, Interest Rate and Exchange Rate data. It shows results from stationarity testing, model performance, most influential variables in both models and also comparison between univariate and multivariate performance of both models.

4.2 Stationarity Testing

The original Public Debt data was tested for stationarity using the Augmented Dickey-Fuller (ADF) test. The results showed a high p-value of 0.976 indicating non-stationarity. To solve this, second order differencing was done to achieve stationarity using the `.diff()` function in Python. Figure 4.1 shows the differenced total debt data over time. Differencing is a technique in time series that helps to transform non-stationary data into a stationary form to stabilize mean and eliminate trends as explained by Verma et.al (2021).

From the results, the preferred level of differencing was the 2nd and was necessary due to the presence of strong trends and non-stationarity in the raw data. From equation (3.5), the ADF test regression for the second-differenced series is specified as:

$$\Delta^2 y_t = \Delta(\Delta y_t) = \Delta y_t - \Delta y_{t-1} \quad (4.1)$$

$$\Delta^2 y_t = (y_t - y_{t-1}) - (y_{t-1} - y_{t-2}) \quad (4.2)$$

$$\Delta^2 y_t = y_t - 2y_{t-1} + y_{t-2} \quad (4.3)$$

Now substituting into the autoregressive framework:

$$\Delta^2 y_t = \alpha + \beta t + \rho y_{t-1}^{(2)} - y_{t-1}^{(2)} + \varepsilon_t \quad (4.4)$$

$$\Delta^2 y_t = \alpha + \beta t + (\rho - 1)y_{t-1}^{(2)} + \varepsilon_t \quad (4.5)$$

Letting $\gamma = \rho - 1$:

$$\Delta^2 y_t = \alpha + \beta t + \gamma y_{t-1}^{(2)} + \varepsilon_t \quad (4.6)$$

Adding lagged differences gives:

$$\Delta y_t^{(2)} = \alpha + \beta t + \gamma y_{t-1}^{(2)} + \sum_{i=1}^p \delta_i \Delta y_{t-i}^{(2)} + \epsilon_t \quad (4.7)$$

where $y_t^{(2)} = \Delta^2 y_t$ denotes the second-differenced public debt series.

The test results showed a p-value of 0.001 which is < 0.05 , leading to rejection of the null hypothesis of a unit root. This confirmed that the second-differenced public debt series is stationary and suitable for ARIMAX modeling.

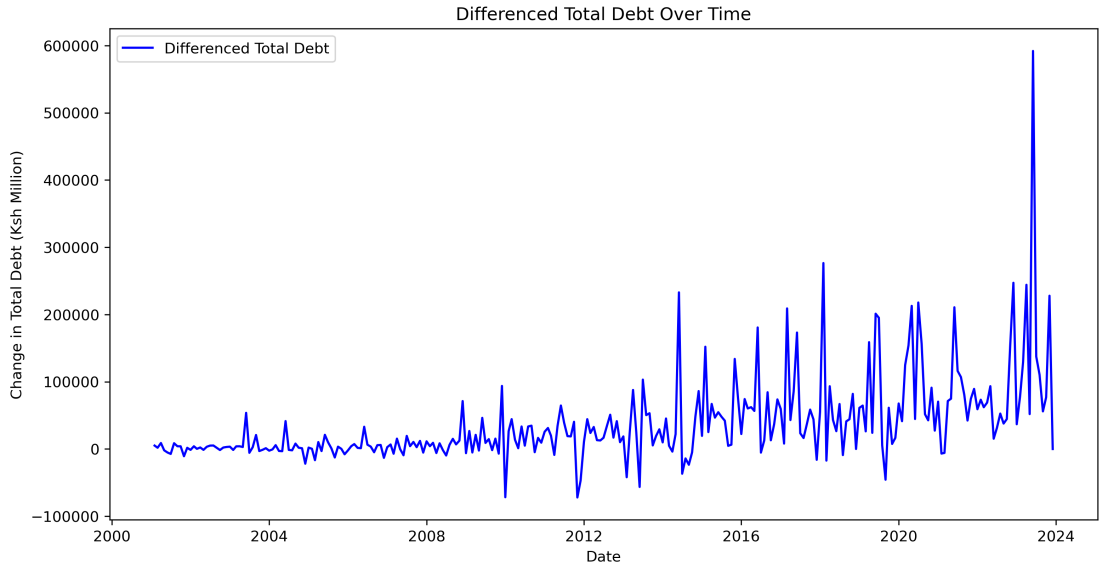


Figure 4.1: Differenced Total Debt over Time

4.3 ARIMAX Model Implementation

An exhaustive Grid Search was conducted to identify the best fitting ARIMAX model with a training set which was 80 percent (2001-2019) of the data. The search explored various combinations of the (p, d, q) parameters and seasonal components (P,D,Q,s) where s represents the seasonal period of 12 months. Akaike Information Criterion (AIC) was used to evaluate the model's performance. The best fitted model had an order of (1,2,2) with first order autoregressive process, second order differencing, and second order moving average and an AIC value of 4663.86. Substituting these values of the optimal model in equation (3.10) gives us:

$$\Delta^2 y_t = c + \phi_1 \Delta^2 y_{t-1} + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \sum_{i=1}^4 \beta_i x_{i,t} + \epsilon_t \quad (4.8)$$

where $\Delta^2 y_t$ is the second differenced public debt at time t , c is a constant term, ϕ_1 is the autoregressive coefficient of order 1, θ_1, θ_2 are the moving average coefficients of orders 1 and 2 respectively, $x_{i,t}$ for $i = 1, \dots, 4$ are the exogenous macroeconomic variables (GDP, Inflation, Exchange Rate, Interest Rate), β_i are the coefficients corresponding to each exogenous variable, and ϵ_t is the white noise error term at time t .

The ARIMAX model, ARIMAX(1,2,2), indicates the following structure:

$p = 1$: A first-order autoregressive term, represented as: $\phi_1 \Delta^2 y_{t-1}$

$d = 2$: A second-order differencing was applied to the series, which is defined as: $\Delta^2 y_t = y_t - 2y_{t-1} + y_{t-2}$

$q = 2$: Two lagged error terms were included, which are: $\theta_1 \epsilon_{t-1}$ and $\theta_2 \epsilon_{t-2}$

Table 4.1: ARIMAX Model Coefficients

Parameter	Variable	Coefficient	Std. Error	p-Value
ϕ_1	Debt_Lag1	0.8031	0.343	0.019
θ_1	MA(1) (ma.L1)	-0.0319	0.336	0.924
θ_2	MA(2) (ma.L2)	-0.1528	0.099	0.122
β_1	GDP	0.0128	0.006	0.021
β_2	Inflation	1040.0073	3276.907	0.751
β_3	Exchange_Rate	64.9272	494.731	0.896
β_4	Interest_Rate	-16340.00	1118.701	0.001

Assuming $c = 0$ since no intercept was listed, equation (4.8) therefore becomes:

$$\begin{aligned} \Delta^2 y_t = & 0.8031 \times \Delta^2 y_{t-1} - 0.0319 \times \epsilon_{t-1} - 0.1528 \times \epsilon_{t-2} \\ & + 0.0128 \times \text{gdp}_t + 1040.0073 \times \text{Inf}_t + 64.9272 \times \text{Ex}_t \\ & - 16340.00 \times \text{Int}_t + \epsilon_t \end{aligned} \quad (4.9)$$

4.3.1 Residual Diagnostics

Four key diagnostic plots were examined to assess the models' assumptions:-

Residual Vs Fitted values: Figure 4.2 shows the residuals to be randomly scattered around the zero line indicating that the model captures systematic variation in the data adequately. There is no clear pattern or trend suggesting that errors are random. This suggests that the model errors are homoscedastic (constant variance) and that the model has not missed any significant non linear relationships in the data. However, there might be a slight suggestion of increasing variability as the spread of residuals increases slightly but this effect isn't pronounced. These observations imply that the linearity assumptions of the ARIMAX model is largely satisfied since residuals fluctuate randomly around the horizontal axis.

Histogram of Residuals: Figure 4.3 demonstrates that the residuals are approximately normally distributed with the distribution appears bell-shaped and centered around zero which aligns with the assumption of unbiased error terms. However, the distribution appears to be positively skewed with coefficient value of 1.9027. Furthermore, residuals display heavy tails as indicated by the excess kurtosis of coefficient 8.58. These features imply that while ARIMAX model captures the central trend of public debt reasonably well, some extreme deviations exist. This could stem from structural breaks or regime shifts.

Autocorrelation of Residuals: Figure 4.4 shows that most of the autocorrelation spikes are within the 95% (± 0.1321) confidence interval. This shows that the residuals are largely uncorrelated suggesting that the ARIMAX model has effectively captured the temporal dependencies and the dynamic relationship between Public Debt and the macroeconomic variables. In other words, there is no significant remaining autocorrelation in the residuals, which supports the reliability and correctness of the fitted model.

Ljung-Box Test was also used to test for autocorrelation. The test results (lb stat = 2.842618, p-value = 0.984889 at lag 10) indicate that there is no statistically significant evidence of autocorrelation since the p-value is greater than the significant level of 0.05.

The residual diagnostics conducted on the ARIMAX model indicate that the model adequately fits the data and meets the assumptions necessary for reliable forecasting.

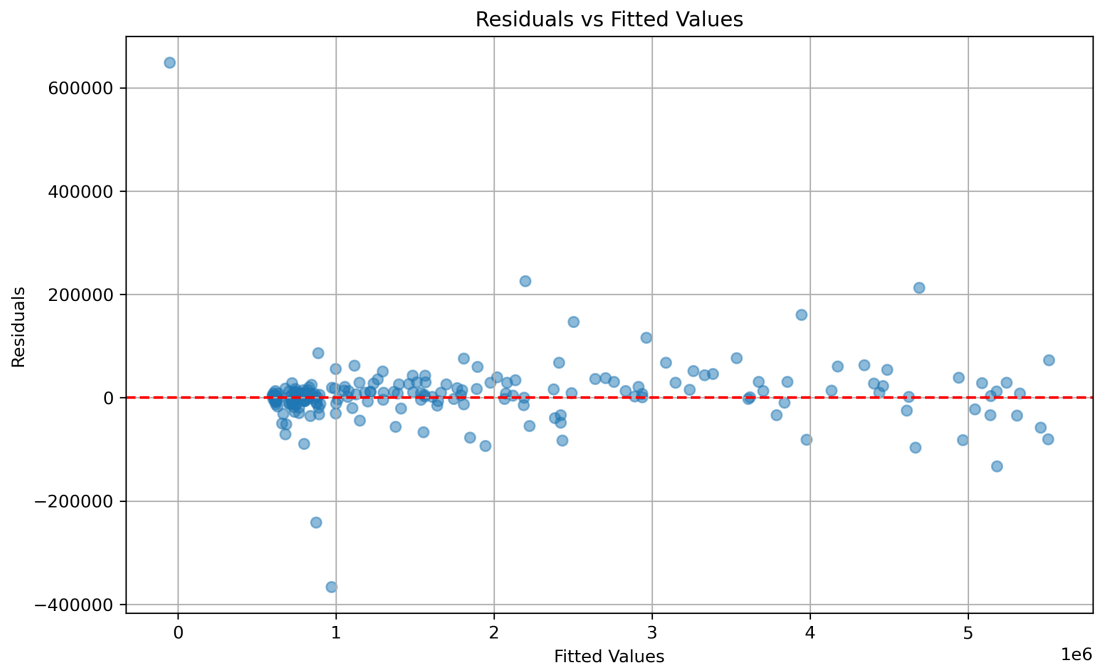


Figure 4.2: Residuals vs Fitted Values

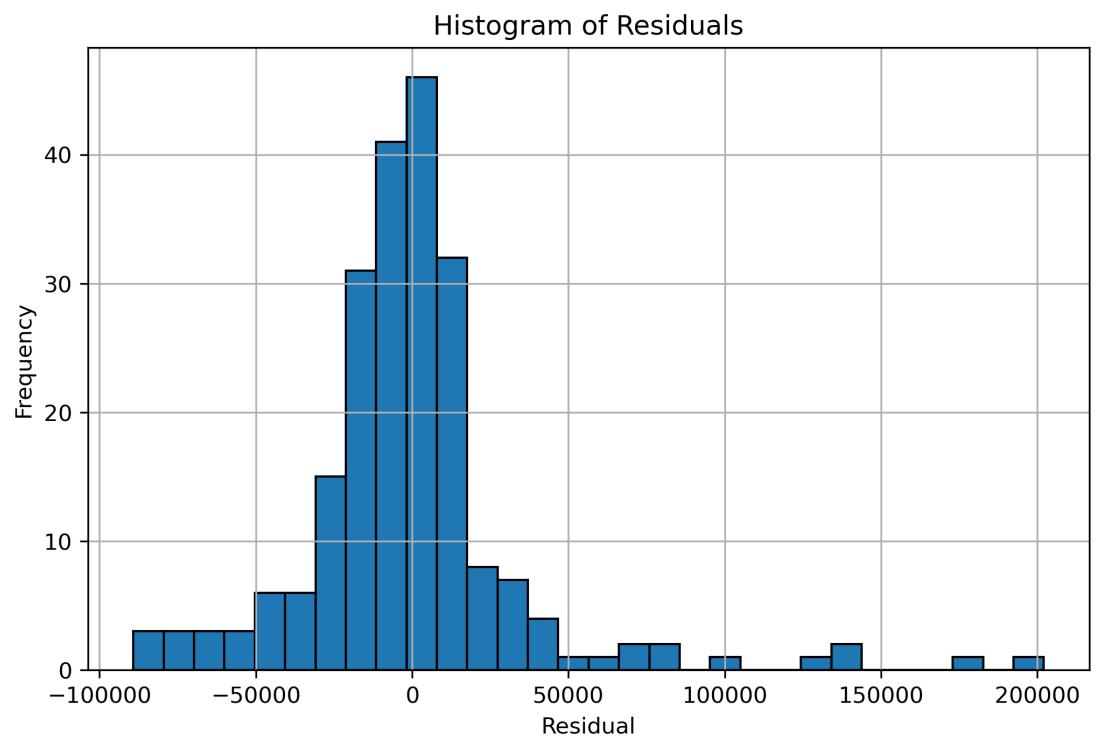


Figure 4.3: Histogram of Residuals

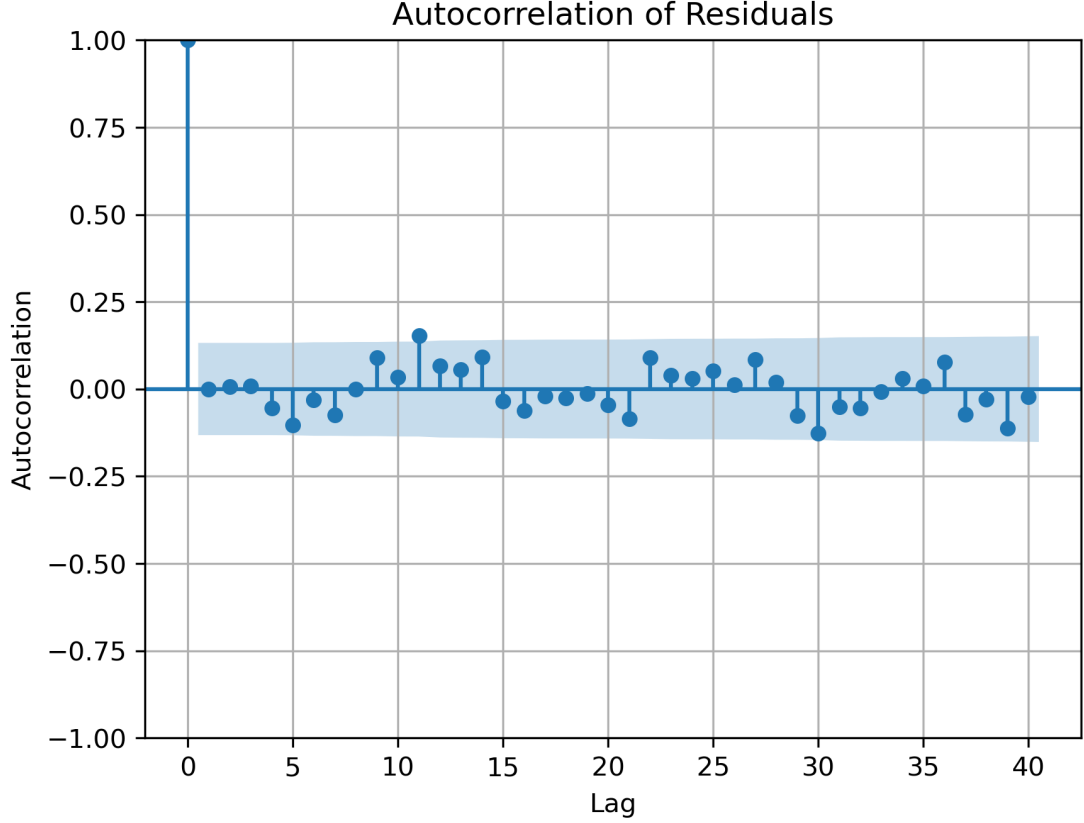


Figure 4.4: Autocorrelation of Residuals

4.4 XGBoost Model Implementation

For this model, lag features were introduced to enable XGBoost to learn from temporal dependencies in the data. Each macroeconomic variable and the target variable, Public Debt, were shifted by one time step (i.e one year) to create lagged versions such as gdp-lag1, inflation-lag1, and so forth. The lag features allow the model to capture the influence of prior values on current debt levels. The use of lagged inputs helps to incorporate time-series dynamics into machine learning models like XGBoost, Rahul Holla, (2024) To optimize the performance of the XGBoost model, a combination of GridSearch and TimeSeriesSplit with 5 fold cross validation was conducted over a range of hyperparameters. The optimal set of hyperparameters were:-Number of Estimators: 300, Learning Rate: 0.01, Max. depth of trees: 3, L1 regularization term: 0.5, and L2 regularization term: 0.5. This set was used to train the final XGBoost model on the full training dataset which was 80% of the data (2001-2019).

The predicted public debt at time t , denoted by $\hat{y}_t$, is given by the sum of outputs from 300 additive regression trees. Substituting equation (3.13) with $K = 300$ gives:

$$\hat{y}_t = \sum_{k=1}^{300} f_k(\mathbf{x}_t), \quad f_k \in \mathcal{F} \quad (4.10)$$

The objective function minimised:

$$\text{Obj} = \sum_{i=1}^{263} l(y_i, \hat{y}_i) + \sum_{k=1}^{300} \Omega(f_k) \quad (4.11)$$

The regularization term, $\Omega(f_k)$ in (4.11), penalizes the complexity of each individual tree and is defined as:

$$\Omega(f_k) = \frac{1}{2} \lambda \sum_{j=1}^{T_k} w_{kj}^2 + \alpha \sum_{j=1}^{T_k} |w_{kj}| \quad (4.12)$$

Therefore, substituting (4.12) in (4.11) gives:

$$\text{Obj} = \sum_{i=1}^{263} l(y_i, \hat{y}_i) + \sum_{k=1}^{300} \left(\frac{1}{2} \times 0.5 \sum_{j=1}^{T_k} w_{kj}^2 + 0.5 \sum_{j=1}^{T_k} |w_{kj}| \right) \quad (4.13)$$

where y_i is the actual public debt at time i , $\hat{y}_i$ is the predicted public debt at time i , f_k represents the k^{th} regression tree in the ensemble, $\mathbf{x}_i$ is the vector of exogenous macroeconomic variables (GDP, Inflation, Exchange Rate, Interest Rate) at time i , 300 is the number of boosting iterations (estimators), T_k is the number of leaves in tree k , w_{kj} is the score on leaf j of tree k , λ and α are the L2 (Ridge) and L1 (Lasso) regularization parameters, respectively, the hyperparameters include learning rate $\eta = 0.01$, max depth $d = 3$, column subsample 0.9, and subsample 0.7, and the gamma parameter $\gamma = 0$ indicates no additional penalty on the number of leaves.

4.5 Influence of Macroeconomic Variables on Public Debt

This study evaluated the relative influence of macroeconomic indicators i.e. GDP, inflation, exchange rate, and interest rate, on Kenya's public debt. Figure 4.3 shows the feature importance derived from the XGBoost model. The horizontal axis (X-axis) lists the 4 macroeconomic variables while the vertical axis (Y-axis) represents the importance score which is calculated by the XGBoost algorithm to quantify how much each feature contributes to the model's predictive accuracy. Higher scores indicate greater importance. Exchange Rate had the highest importance score, suggesting it had the highest influence on public debt. GDP was the second most important feature followed by Inflation. They both played a noticeable role in the model's predictions but their effect was far less than Exchange Rate's. Interest Rate was the least important meaning it had the smallest impact on predictions compared to the other features.

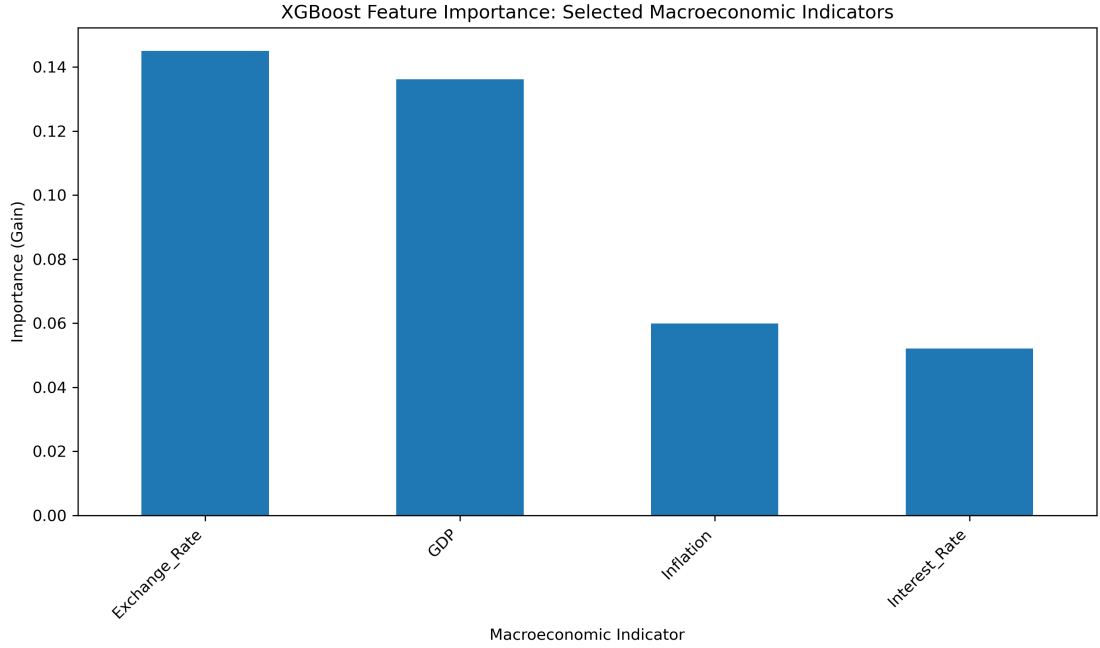


Figure 4.5: Feature importance from XGBoost model

Table 4.2 presents the ARIMAX model results. Notably, interest rate had a statistically significant negative relationship with public debt (coefficient = -16,340; p-value = 0.001), suggesting that increases in interest rates may reduce debt accumulation. In contrast, GDP has a small but statistically significant positive effect (coefficient = 0.0128; $p = 0.021$) implying that economic growth is associated with a marginal increase in public debt. Inflation and Exchange Rate were not statistically significant $p > 0.05$, indicating weak influence.

Table 4.2: ARIMAX Model Coefficients for Exogenous Variables

Variable	Coefficient	p-value	Significance
Exchange Rate	64.93	0.896	Not Significant
Inflation	1040.01	0.751	Not Significant
GDP	0.0128	0.021	Significant
Interest Rate	-16,340	0.001	Significant

In summary, Exchange Rate emerged as the most influential predictor based on machine learning (XGBoost), while Interest Rate had a statistically significant impact based on the traditional ARIMAX model. This contrasting outcomes in variable importance between the 2 models can be as a result of their different modeling approaches. ARIMAX identifies linear effects that are statistically significant, $p < 0.05$, while XGBoost, which is capable of modeling non-linearities, might find variables with strong non-linear relationships to be important predictors. ARIMAX tells more about the statistically significant linear impact of each variable on the target variable, public debt, while XGBoost tells more about the predictive power of each variable (including lagged versions) within a complex, non-linear model. Therefore, it is possible for a variable to

have a statistically significant linear effect in ARIMAX model but have relatively low feature importance in XGBoost if its relationship with the target variable is primarily linear.

4.6 Determining the Optimal Model for Forecasting Kenya's Public Debt

The ARIMAX and XGBoost models were compared to determine the most effective model to forecast public debt in Kenya using GDP, Inflation, Interest Rate and Exchange Rate as macroeconomic variables. Forecasting for both models was done for a period of 4 years (i.e 2020-2023). The performance of each model is evaluated based on visual inspection of the predicted and actual debt trajectories and accuracy measures such as Mean Absolute Percentage Error(MAPE), Root Mean Square Error (RMSE), and R-Squared were used to compare the forecasting accuracy for each model.

4.6.1 Comparative Evaluation of both ARIMAX and XGBoost Models

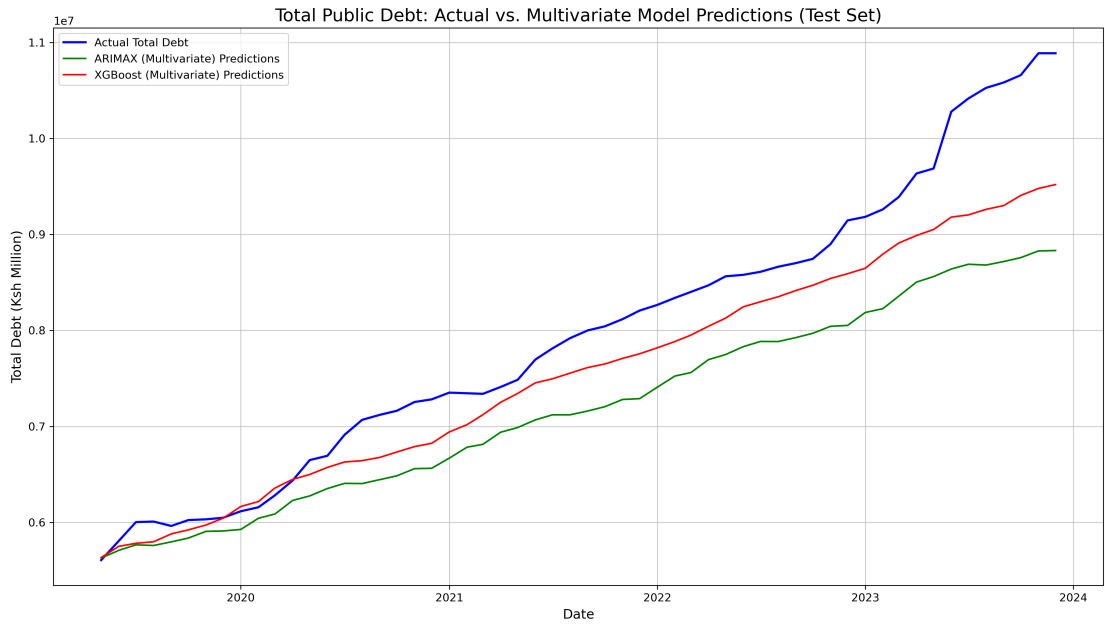


Figure 4.6: Actual Vs Predicted Public Debt (ARIMAX and XGBoost With Macroeconomic Variables)

Figure 4.6 shows the performance of ARIMAX and XGBoost models in predicting the total public debt in Kenya over the test period of 2020-2023. The X-axis represents the date in years while the Y-axis represents public debt in units of 10 million. The blue line represents the actual public debt, green line represents the predicted values from ARIMAX model and the red line shows the predictions from XGBoost model. The 2 models capture the upward trend of the

actual public debt. However, the degree to which they follow the nuances of the actual data varies.

The red line representing XGBoost predictions appears to track the actual public debt (blue line) more closely than the ARIMAX predictions. The model demonstrates a superior ability to capture underlying patterns and turning points in the actual public debt. The non-linear nature of the model likely allows it to adapt more effectively to the complexities of debt accumulation. The green line representing ARIMAX predictions also appears to be capturing the general upward trend but is less responsive to the dynamic changes in the rate of debt accumulation and tends to lag behind the actual public debt. The ARIMAX model struggles to fully capture the non linear accelerations in debt accumulation.

Based on accuracy measures, Table 4.3 shows how each model performed. XGBoost had a lower RMSE value of 556,672.24 compared to ARIMAX's 1,030,374.56 confirming that on average, its predictions are more accurate than ARIMAX. XGBoost's MAPE value was 4.84% while ARIMAX had 9.52% showing its predictions are more accurate in percentage terms. The R-squared value was 0.86 compared to ARIMAX's 0.51 suggesting that XGBoost explains a much larger portion of the variability in the actual public debt.

Table 4.3: Performance Metrics of Forecasting Models

Model	RMSE	MAPE (%)	R-squared
XGBoost	556,672.24	4.84	0.86
ARIMAX	1,030,374.56	9.52	0.51

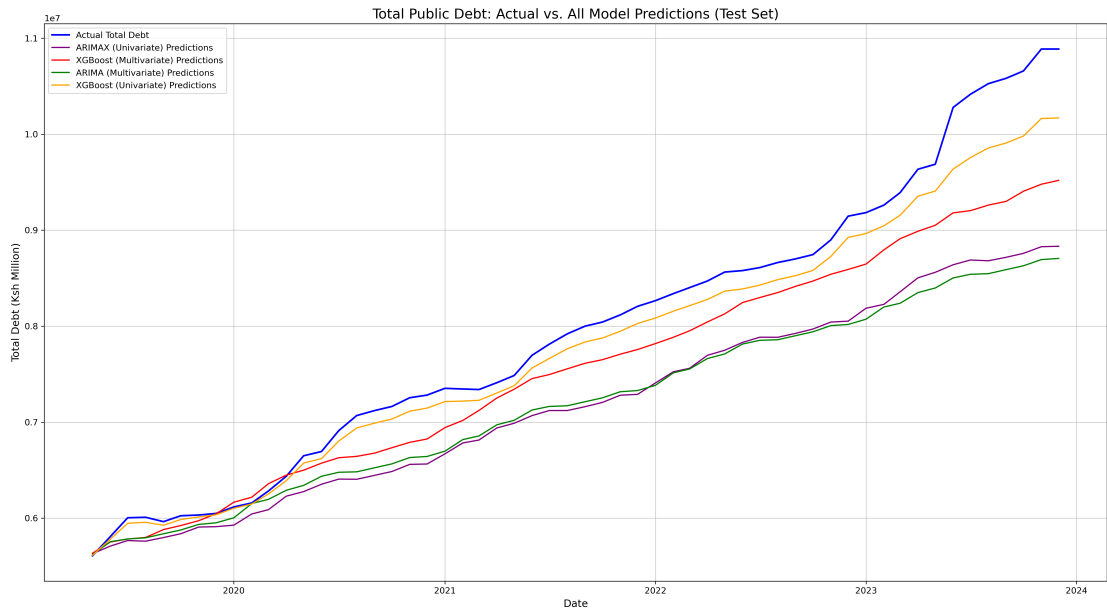


Figure 4.7: Actual Vs Predicted Public Debt for both Univariate and Multivariate Models

Table 4.4: Comparison of Actual and Predicted Yearly Total Public Debt (in KSH Millions)

Year	Actual Debt	ARIMAX Multi	XGBoost Multi	ARIMA Uni	XGBoost Uni
2019	47.50	46.32	46.79	46.55	47.26
2020	81.14	75.78	78.56	76.74	80.11
2021	92.73	84.44	88.91	84.94	91.02
2022	103.38	93.53	98.74	93.22	101.16
2023	121.39	102.98	109.74	101.47	115.40

Figure 4.7 shows the performance of Univariate and Multivariate models over the test period of 2020-2023. The ARIMA model (purple line) achieved an RMSE of 960,773.95, a MAPE of 8.78%, and an R^2 of 0.57. These results indicate that while ARIMA captured the trend to some extent, its accuracy and explanatory power were moderate. The ARIMA model is inherently linear and may not effectively handle non linear dynamics in the data. In contrast, the univariate XGBoost model (orange line) outperformed ARIMA with an RMSE of 278,627.53, a MAPE of 2.23%, and an R^2 of 0.96. These results demonstrate exceptional forecasting accuracy. XGBoost's ability to model complex non linear relationships and interactions within a time series allows it to detect subtle trends and seasonality that linear models may overlook. This observation aligns with findings by Yu et al. (2019), who found that tree based ensemble methods like XGBoost can outperform traditional statistical models in time series forecasting tasks due to their robustness, flexibility, and resistance to overfitting.

The XGBoost multivariate model (red line) , while still performing well, yielded a higher RMSE (556,672.24) and MAPE (4.84%) compared to the univariate model. The R^2 slightly decreased to 0.86, suggesting that while the model still explained nearly all the variance in public debt, its relative prediction error increased with the inclusion of additional variables. Similarly, the ARIMAX model (green line), which extends ARIMA by incorporating exogenous variables, performed poorly with an RMSE of 1,030,374.56, a MAPE of 9.52%, and an R^2 of only 0.51. These results indicate that the inclusion of macroeconomic variables did not substantially enhance the forecasting power. Zhang et al. (2018) emphasized that multivariate models do not always outperform univariate ones, especially when the external variables are weakly correlated with the target variable or when their relationships are non-linear and poorly captured by the model. Overfitting or multicollinearity can also diminish the model's generalizability and prediction accuracy.

Table 4.5: Performance Metrics of Models for Univariate and Multivariate Data

Model	Univariate			Multivariate		
	RMSE	MAPE (%)	R^2	RMSE	MAPE (%)	R^2
ARIMA	960,773.95	8.78	0.57	-	-	-
XGBoost	278,627.53	2.23	0.96	556,672.24	4.84	0.86
ARIMAX	-	-	-	1,030,374.56	9.52	0.51

4.7 Future Predictions for XGBoost and ARIMAX Optimal Models

Figure 4.8 illustrates the future projections of Kenya’s total public debt from 2024 to 2028 using the optimal ARIMAX and XGBoost models, alongside the historical trend.

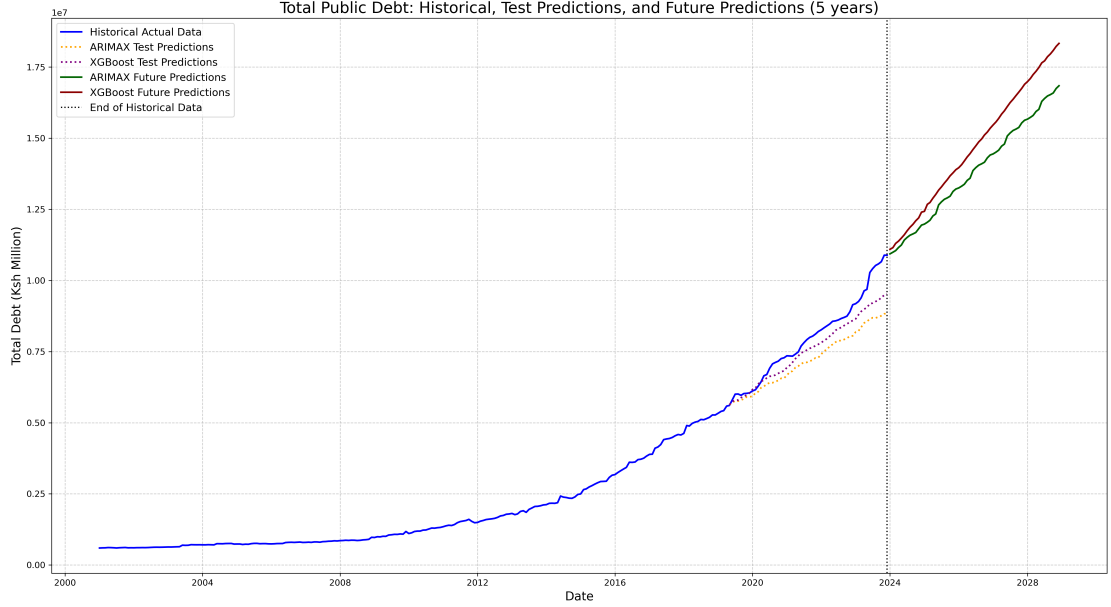


Figure 4.8: Future Predictions plot for XGBoost and ARIMAX Optimal models

As visualized, the ARIMAX model’s test predictions (orange dashed line) and XGBoost’s (purple dashed line) both followed the historical upward trajectory of Kenya’s public debt. While the models captured the overall trend, they slightly underestimated the sharp increase in debt observed during the latter part of the test period, particularly from early 2022 to late 2023. This underperformance during periods of rapid change is common in forecasting, as models trained solely on prior data may lag behind recent shifts as discussed by Hyndman Athanasopoulos, (2018).

For the subsequent 5 year forecast, both models were retrained on the entire historical dataset spanning from 2001 to 2023. This retraining process is standard in time series forecasting and ensures that the models incorporate the most recent information to produce more robust forward-looking estimates Makridakis, Spiliotis, Assimakopoulos, (2018). After retraining, both models generated future projections (solid lines) that diverged noticeably from their earlier test predictions. For ARIMAX, the solid green future projection line indicated a much steeper rise in debt compared to its previous test curve. This divergence is not a contradiction but rather a reflection of the model’s sensitivity to recent trends. The ARIMAX model learns and extrapolates temporal patterns, and since the recent data included a pronounced acceleration

in debt growth, the model projected this trend forward more aggressively as supported by the findings of Box, Jenkins, Reinsel, (2015).

Similarly, the XGBoost model (dark red solid line) projected a significant rise in public debt. Despite their methodological differences, ARIMAX being a statistical model with explicit temporal structures, and XGBoost being a tree-based machine learning algorithm, both models produced future trajectories that suggested a continued, strong upward momentum in public debt. This consistency reinforces the reliability of the projection and demonstrates that both modeling approaches have internalized the recent trend patterns. The test predictions are based on models trained without access to the test set, meaning they perform interpolation within a limited historical context. In contrast, the future projections are extrapolations informed by the full dataset, including the most recent test period trends. Therefore, the retrained models are expected to show stronger growth patterns, having “learned” the late stage acceleration that was absent from the earlier training window Tashman, (2000). This methodological distinction helps explain the visual difference between the dashed and solid lines in the forecast graphs.

These results also highlight model responsiveness to recent data. ARIMAX, which models time-dependent relationships explicitly, responded sharply to the sudden increase in recent debt, while XGBoost, with its non linear structure, was equally able to capture the upward trend. Retraining on the full data proved essential in both cases. Including the most recent observations enabled the models to adjust to the current fiscal reality, leading to more realistic forecasts. This outcome underscores the value of regularly updating forecasting models to maintain accuracy, particularly in volatile macroeconomic settings.

Table 4.6: Forecasted Public Debt in Kenya (2024–2028) using ARIMAX and XGBoost Models

Year	ARIMAX (Million KSH)	XGBoost (Million KSH)
2024	136.98	140.30
2025	151.21	158.53
2026	165.90	175.87
2027	180.44	194.02
2028	195.02	211.97

4.8 Scenario Simulation

The graphs below show the simulation of public debt, debt-to-GDP, debt-to-Exchange Rate (XGBoost), and debt-to-Interest Rate (ARIMAX) under Baseline, Optimistic, and Pessimistic scenarios. The 3 scenarios incorporated assumptions about macroeconomic trends. They were modeled using a combination of linear and compound growth approaches, and adjusted using model specific amplification factors based on variable importance. The forecasting horizon ex-

tends for 60 months (5 years) from the last observed data point, with each forecasted variable initialized using its final historical value. These served as baseline reference points for forward projections under each scenario. To introduce structural variability over time, the 5 year forecast horizon was split into two equal phases. Phase 1 covered the first 2.5 years, and Phase 2 covered the next 2.5 years. This phased structure allowed for piecewise changes, where macroeconomic indicators evolved differently across the two sub-periods, thus capturing potential mid-horizon shifts in the economic environment.

The baseline scenario assumed a stable economic outlook with gradual and moderate changes across all exogenous variables. GDP was projected to grow by 10% over the 5 year period, modeled as compound growth equivalent to a 0.04 rate scaled to the period length, yielding an approximate annual growth rate of 1.92%. Inflation was expected to decline linearly by 1.0 percentage point, while the exchange rate appreciated by 12.5% compound over the same period (equivalent to a 0.05 rate scaled to 5 years). The interest rate was projected to decrease linearly by 1.0 percentage point, reflecting easing borrowing conditions.

The Optimistic Scenario assumed improved economic conditions and stronger macroeconomic performance. GDP grew in a piecewise compound manner 5% in Phase 1 and 1.25% in Phase 2. Inflation decreased by 1.5 percentage points during Phase 1 but rebounded slightly by 0.5 points in Phase 2. The exchange rate appreciated by 3.75% in Phase 1 and 1.25% in Phase 2 (compound). Interest rates were projected to decline by 1.5 percentage points in Phase 1, followed by a slight increase of 0.5 points in Phase 2. The overall economic narrative in this scenario was one of strong early gains, followed by moderate but sustained stability.

The Pessimistic Scenario simulated a more challenging economic environment. GDP experienced very slow compound growth, only 0.625% in each of the two phases, reflecting near stagnation. Inflation increased by 0.75 percentage points in Phase 1 and a further 0.25 points in Phase 2, indicating mounting inflationary pressure. Exchange rates deteriorated, with a compound depreciation of 10% in Phase 1 and 5% in Phase 2. Interest rates followed a similar negative trajectory, rising by 0.75 points in the first half and 0.25 in the second half of the projection period. This scenario reflected a prolonged period of economic strain with weakening currency and rising inflation and borrowing costs.

To better differentiate the optimistic and pessimistic scenarios from baseline, amplification factors were applied to the most significant variables identified within each model. A multiplicative factor of 1.8 was used in these scenarios, meaning that variables with the highest influence underwent 80% more pronounced changes than the baseline projection.

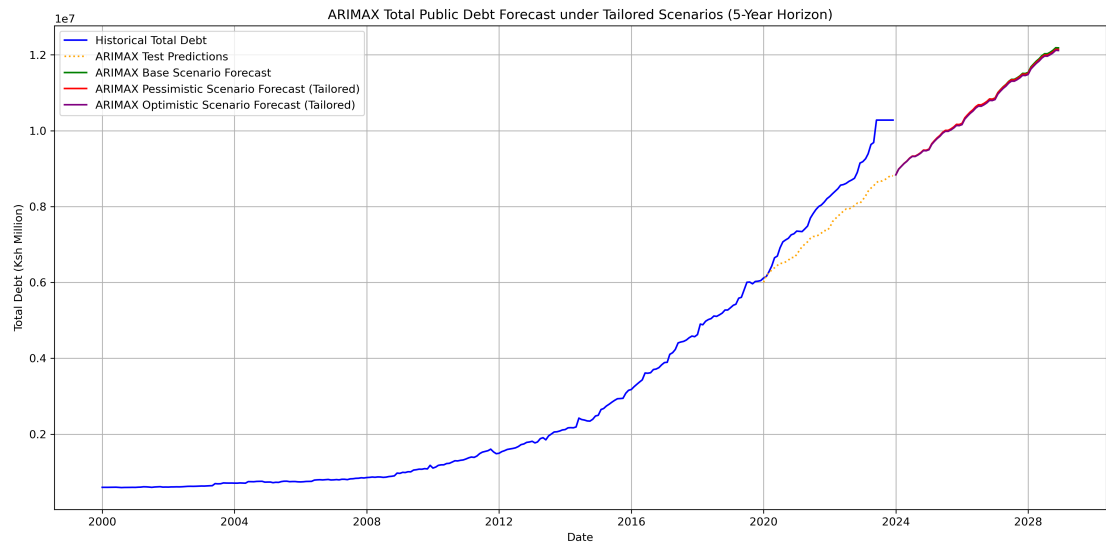


Figure 4.9: ARIMAX Total Debt Scenarios

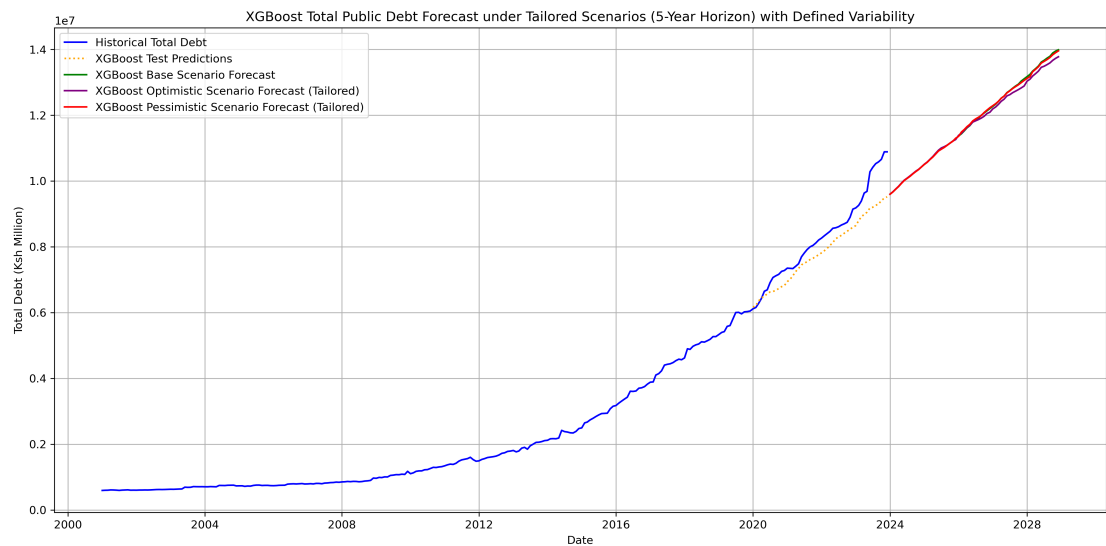


Figure 4.10: XGBoost Total Debt Scenarios

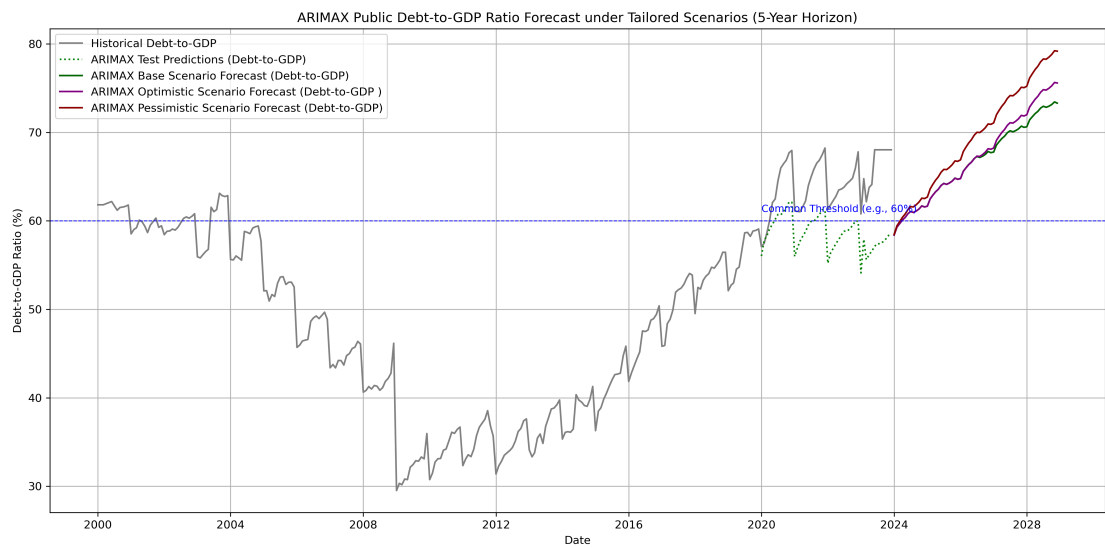


Figure 4.11: ARIMAX Debt-to-GDP Ratio

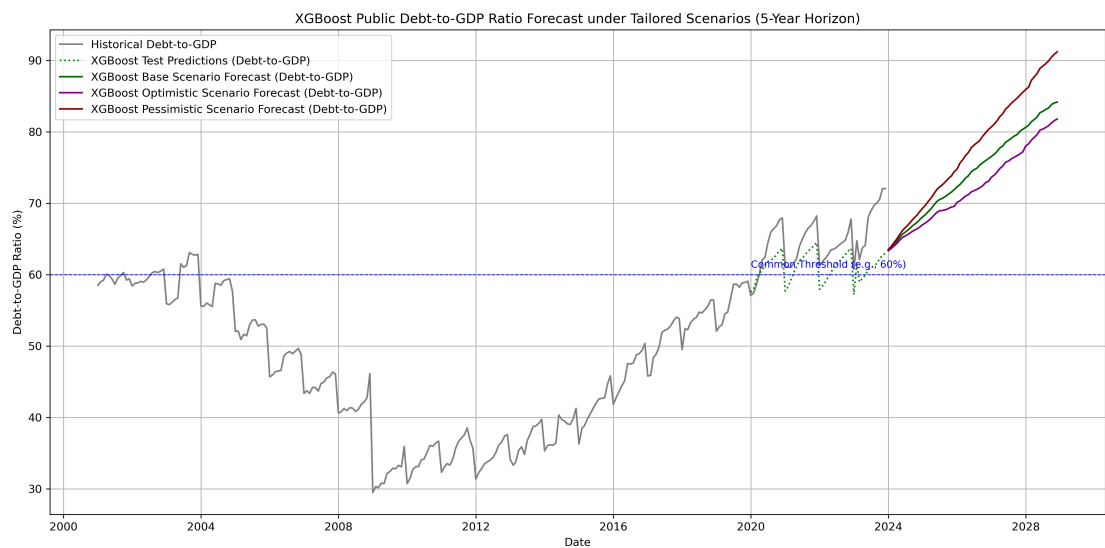


Figure 4.12: XGBoost Debt-to-GDP Ratio

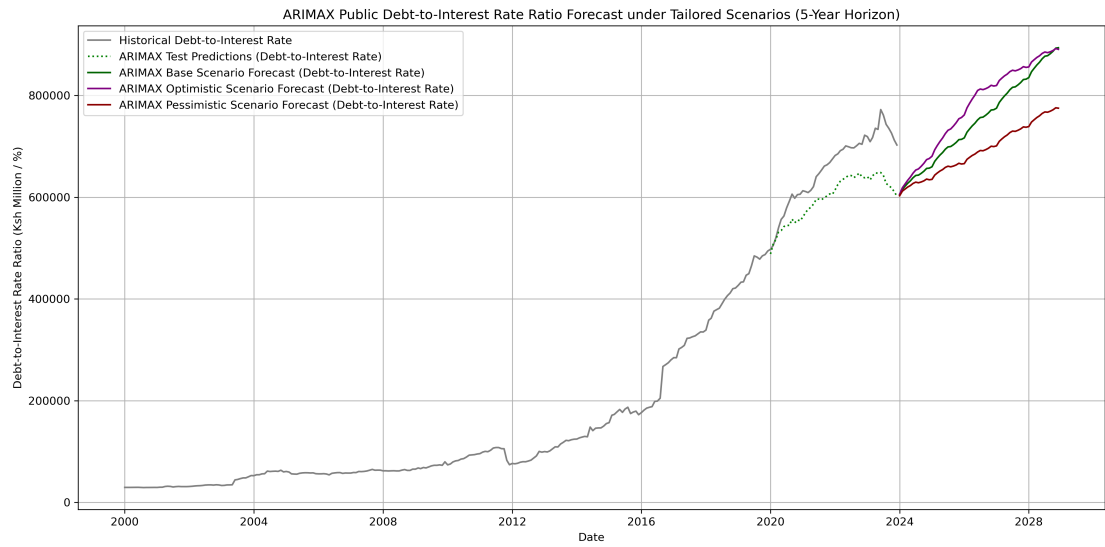


Figure 4.13: ARIMAX Debt-to-Interest Rate Ratio

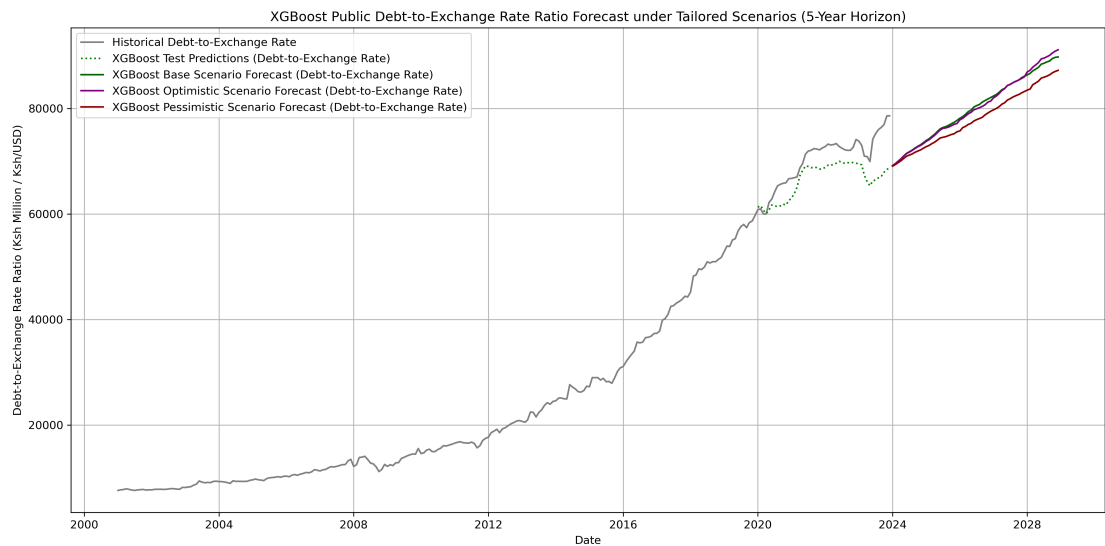


Figure 4.14: XGBoost Debt-to-Exchange Rate Ratio

Table 4.7: Forecasted Total Public Debt under Different Scenarios

Model	Baseline Scenario	Optimistic Scenario	Pessimistic Scenario
XGBoost	13,987,372.54	13,776,184.49	13,954,764.39
ARIMAX	12,182,814.66	12,139,780.18	12,113,112.34

Table 4.8: Forecasted Debt-to-GDP Ratio under Different Scenarios

Model	Baseline Scenario	Optimistic Scenario	Pessimistic Scenario
XGBoost	84.16%	81.81%	91.22%
ARIMAX	73.30%	75.58%	79.18%

Table 4.9: Forecasted Debt-to-Interest Rate Ratio (ARIMAX) under Different Scenarios

Model	Baseline Scenario	Optimistic Scenario	Pessimistic Scenario
ARIMAX	893,823.53	890,666.19	774,991.19

Table 4.10: Forecasted Debt-to-Exchange Rate Ratio (XGBoost) under Different Scenarios

Model	Baseline Scenario	Optimistic Scenario	Pessimistic Scenario
XGBoost	89,777.02	91,133.77	87,241.30

While both models show similar directional trends, the magnitude of differences across the Baseline, Optimistic, and Pessimistic scenarios remained relatively small in most cases. This suggests that the models, especially over a five year horizon, may exhibit limited sensitivity to short term macroeconomic shocks. In terms of total public debt, the XGBoost model forecasted approximately Ksh 13.98 million under the Baseline Scenario, 13.77 million under the Optimistic Scenario, and 13.95 million under the Pessimistic Scenario. ARIMAX showed a slightly lower range, projecting 12.18 million for the Baseline, 12.14 million for Optimistic, and 12.11 million for Pessimistic. The relatively minor differences across the scenarios imply that neither model anticipates drastic shifts in public debt accumulation in response to the simulated macroeconomic paths. A possible explanation is that the compounding or linear changes introduced in the exogenous variables, even when amplified, may have been too moderate to elicit large variations in public debt. Moreover, ARIMAX's linear structure tends to smooth out volatility, and while XGBoost captures non linear interactions, it may also reflect offsetting effects among variables like GDP, exchange rate, and inflation.

When evaluating debt-to-GDP ratios, the models provided a more intuitively aligned outcome, particularly under the Pessimistic Scenario. In XGBoost, the ratio rose from 84.16% under the Baseline to 91.22% under Pessimistic conditions. Similarly, in ARIMAX, it increased from 73.30% to 79.18%. This reflected the reality that slower GDP growth, as simulated under adverse conditions, weakens the denominator in the ratio, resulting in a higher debt burden relative to economic output. The scenario based forecasts from the XGBoost and ARIMAX

models provide important insights into the potential trajectory of Kenya’s public debt under varying macroeconomic conditions. However, in the Optimistic Scenario, the debt-to-GDP ratio was slightly higher than in the Baseline for ARIMAX model, 75.58%. This counterintuitive result was maybe due to the optimistic growth in GDP being insufficient to offset debt growth in the short run, especially when modeled with lag structures or moderate compounding.

The debt-to-interest rate ratio under the ARIMAX model, contrary to expectation, showed the lowest ratio under the Pessimistic Scenario, at approximately 774,991, compared to 893,823 in the Baseline and 890,666 in the Optimistic case. It is generally expected for this ratio to rise under pessimistic conditions due to increasing debt and higher interest rates. However, the result is mathematically justified when considering the structure of the ratio. Since it is computed as public debt divided by the interest rate, a significant increase in the denominator (interest rate) combined with relatively stable debt leads to a lower ratio. This does not indicate improved debt sustainability but rather reflects the mechanical effect of an increasing denominator. A similar explanation applies to the debt-to-exchange rate ratio in XGBoost. The lowest ratio is observed under the Pessimistic Scenario (87,241), while the highest is under the Optimistic Scenario (91,133). Again, although this appears contradictory, it is consistent with the ratio structure, where currency appreciation in the Optimistic Scenario decreases the denominator less than debt declines, resulting in a higher ratio, and vice versa under depreciation.

4.9 Discussion

The results from this study reveal significant differences in the forecasting performances of ARIMAX and XGBoost models when applied to Kenya’s public debt data. XGBoost outperformed ARIMAX across all evaluation metrics achieving lower RMSE and MAPE, and a higher R^2 value. The poor performance of ARIMAX, with an RMSE of 1,030,374.56 and R^2 of 0.51, is evidence of the limitations of traditional time series models in capturing non-linear relationships and structural changes that are often present in macroeconomic data. Hyndman and Athanasopoulos, (2018) reported similar findings where they noted that ARIMA based models often underperform in the presence of highly volatile or shifting economic series. XGBoost, a powerful ensemble learning technique based on gradient boosting, demonstrated excellent predictive performance with a MAPE of 4.84% and a near perfect R^2 of 0.86. This aligns with prior research by Zhang et al., (2019) and Makridakis et al., (2018) who found that machine learning models like XGBoost and Random Forests consistently outperform statistical models in time series forecasting tasks, particularly when non linearity and outliers are present.

However, the performance dynamics shift when comparing univariate models to their multivariate counterparts. In the univariate context, where only public debt data is used without the inclusion of macroeconomic variables, XGBoost significantly outperformed ARIMA, achieving an RMSE of 278,627.53 a MAPE of 2.23%, and an R^2 of 0.96. ARIMA, on the other hand, had a higher RMSE of 960,773.95 and a higher MAPE of 8.78%, and a lower R^2 of 0.57. These results challenge the conventional assumption that statistical models like ARIMA are superior in univariate settings due to their ability to model autocorrelation and trend components. Instead, they highlight how modern machine learning approaches, when supported with effective feature engineering, can capture temporal dependencies with greater precision, even in the absence of external variables.

The benefits of multivariate modeling become even clearer with XGBoost, which continued to perform robustly upon the inclusion of macroeconomic indicators such as inflation, GDP, exchange rate, and interest rate. While its performance slightly declined compared to the univariate model, it still maintained strong accuracy with an RMSE of 556,672, a MAPE of 4.84%, and an R^2 of 0.86, demonstrating XGBoost's capacity to model complex, non linear relationships across multiple features. ARIMAX, while improving slightly over its univariate version (RMSE of 922,942.10, MAPE of 8.89%, R^2 of 0.60), still lagged behind XGBoost in predictive accuracy. The relatively weaker performance of ARIMAX may be attributed to the limitations of its linear structure, which struggles to capture non linear relationships between public debt and macroeconomic variables. In contrast, XGBoost's flexible, tree based framework can accommodate complex interactions and non linear patterns, giving it a distinct advantage in multivariate settings. Ultimately, these findings underscore the superior predictive performance and adaptability of machine learning models like XGBoost, both in univariate and multivariate scenarios, when appropriately tuned and engineered.

Furthermore, the ability of XGBoost to handle multivariate input features without strong assumptions about the data makes it well suited for financial and macroeconomic forecasting, Naseem et al., (2021). However, while XGBoost provides high accuracy, it lacks interpretability that ARIMAX offers. ARIMAX models quantify the relationship between independent macroeconomic variables and the dependent variable (Public Debt), making them useful in policy analysis and econometric modelling, Gujarati & Porter, (2009). ARIMAX is therefore valuable where understanding causality and variable effects is more important.

In terms of interpretation of Influential Variables, XGBoost model identified Exchange Rate as the most important macroeconomic determinant of public debt, supporting economic theory and empirical literature suggesting that fluctuations in exchange rates significantly affect a country's

external debt servicing costs, especially for economies that rely heavily on foreign-denominated loans, Jongwanich (2009). A depreciation of the domestic currency increases the local currency value of external debt, thereby raising the overall debt burden. This dynamic often leads to higher borrowing needs and accumulating public debt to finance the increased cost of servicing. Machine learning models like XGBoost are well-equipped to capture such nonlinear and interacting effects, making it possible to detect the dominant role of exchange rate variability even when traditional models may overlook it, Zhou et al., (2021). The model's ability to identify this key relationship highlights the importance of exchange rate stability in managing public debt in developing economies like Kenya. On the other hand, ARIMAX indicated a statistically significant negative effect of interest rate on public debt. This aligns with the findings of Reinhart and Rogoff, (2010), who suggested that higher interest rates reduce debt accumulation by increasing the cost of borrowing. Together, the results underscore the complex interplay between macroeconomic variables and public debt. While ARIMA provides robust univariate forecasting when limited data is available, machine learning models like XGBoost offer considerable advantages in multivariate settings by capturing non-linear interactions and improving forecast accuracy.

Applying the optimal models for future predictions, ARIMAX model predicted a sharper, exponential-like growth in public debt, suggesting that the combination of lagged debt values and macroeconomic indicators, possibly under an assumption of continuing unfavorable trends, leads to a compounding effect in forecasted debt levels. In contrast, the XGBoost model produced a more conservative, linear growth trajectory as shown in Figure 4.3. This reflects the model's strong ability to generalize patterns and possibly smooth out short-term fluctuations due to its regularization and tree-based structure. Notably, XGBoost's predictions may have been influenced more heavily by the exchange rate, which emerged as the most influential predictor in its feature importance analysis. This aligns with past research highlighting the exchange rate as a critical driver of external debt servicing and fiscal pressure in developing economies, Jiang et al., (2020). Overall, the divergence between the two models' forecasts highlights the importance of modeling assumptions. While ARIMAX projects higher risk of debt escalation, XGBoost suggests a more stable but steady increase, providing a valuable basis for scenario based policy planning.

In the application of scenario simulation, the contrast in forecasts between the XGBoost and ARIMAX models provides important insights. The XGBoost model, which is capable of capturing non linear interactions among macroeconomic indicators, predicted a sharp rise in both total debt and debt-to-GDP ratio across all scenarios. This could reflect the model's sensitivity to recent trends and patterns in the data, as it prioritizes performance over interpretability as

explained by Varian, (2014). However, such models may be prone to overfitting, especially in volatile economic contexts. The high debt-to-GDP ratio forecasts from XGBoost, exceeding 80% in all scenarios, raise significant concerns about debt sustainability. According to the International Monetary Fund (IMF) (2023), developing economies should aim to maintain this ratio below 60% to ensure fiscal sustainability and avoid triggering debt distress. Therefore, the XGBoost projections suggest that, if current trends persist, Kenya may be heading into a high-risk fiscal zone. On the other hand, the ARIMAX model, grounded in classical time series econometrics, displayed a more restrained forecast pattern. Its predictions of declining total debt and debt-to-GDP ratios under all scenarios suggest a possible reversal of the upward debt trend. ARIMAX models emphasize lagged relationships and the structural dynamics of economic indicators, making them potentially more interpretable and stable for economic policymaking as explained by Box et al., (2015). The debt-to-interest rate and debt-to-exchange rate ratios were lower in the pessimistic scenario. This is mathematically consistent, as both interest rates and exchange rates were assumed to rise under pessimistic conditions, increasing the denominator in each ratio and thus lowering the overall value. These outcomes highlight the importance of interpreting ratio-based indicators cautiously, particularly when the denominator is an exogenous variable that shifts significantly across scenarios. Overall, the results underscore the models' responsiveness to economic context and the value of scenario analysis in stress-testing fiscal outlooks.

Scenario based forecasting, used in this study, is particularly crucial in the face of macroeconomic uncertainty. It allows planners to simulate the fiscal impact of optimistic and pessimistic economic paths, providing a range of policy-relevant forecasts OECD, (2020). Together, these results highlight a trade off between predictive performance and interpretability. While machine learning models like XGBoost can detect complex interactions, their forecasts must be carefully interpreted within the broader macroeconomic context.

Chapter 5

Conclusions and Recommendations

5.1 Conclusions

This study employed 2 distinct modelling techniques, ARIMAX and XGBoost, to forecast Kenya's public Debt using macroeconomic variables: GDP, Inflation, Interest Rate and Exchange Rate, spanning from 2001-2023. ARIMAX, which accounts for exogenous variables, offered interpretable linear relationships and statistically significant coefficient estimates. XGBoost, a non-linear tree-based machine learning model, demonstrated strong predictive performance through its ability to capture complex interactions among variables and handle multicollinearity. Both models were evaluated using RMSE, MAPE, R^2 as the evaluation metrics. XGBoost generally outperformed ARIMAX, however, ARIMAX provided more interpretable coefficients and inference on statistical significance as shown on Table 4.1, making it valuable for policy focused insights.

The analysis of variable importance revealed varying perspectives on variable importance. The ARIMAX model, focusing on linear relationships and statistical significance, highlighted GDP and Interest Rate as significant predictors, while Exchange Rate and Inflation showed weaker linear effects. XGBoost model, capable of capturing nonlinear dynamics and assessing overall predictive power, identified Inflation and Exchange Rate as the most influential, with GDP and Interest Rate exhibiting lower importance. This difference underscores the critical role of the chosen modeling framework in understanding the drivers of public debt. ARIMAX model suggests that, in a direct, statistically testable linear manner, GDP and Interest Rate have a significant impact on public debt levels. Conversely, XGBoost model emphasizes the predictive

power of Inflation and Exchange Rate, potentially capturing more complex, indirect, or lagged relationships that might not be evident in a purely linear model. The greater importance of Inflation in the XGBoost model could reflect its broader macroeconomic implications, influencing government spending, and nominal debt values in ways that a linear model might not capture. Similarly, the Exchange Rate's high significance in the non-linear model might stem from relationships that can be non-linear and interactive such as connections to trade balances, capital flows, and the valuation of foreign-denominated debt. While linear models like ARIMAX provide insights into direct, statistically significant effects, non-linear models like XGBoost can uncover potentially more powerful predictive relationships. The choice of model and the interpretation of variable importance should therefore be guided by the specific research question and the underlying assumptions about the nature of the relationships between macroeconomic factors and public debt. Overall, this study confirms that machine learning models, particularly XGBoost, provide an alternative to traditional econometric models for forecasting public debt especially in developing countries like Kenya, where there is a complex economic environment.

This study employed both XGBoost and ARIMAX models to forecast Kenya's public debt under three macroeconomic scenarios. The findings demonstrated the utility of both ARIMAX and XGBoost models in forecasting public debt under varying macroeconomic conditions. While total debt projections across scenarios remained relatively close, the derived indicators, such as debt-to-GDP, debt-to-interest rate, and debt-to-exchange rate ratios, provided more nuanced insights into debt sustainability. The models responded logically to changes in exogenous variables, with the pessimistic scenario consistently reflecting higher risk to debt sustainability, particularly through increased debt-to-GDP ratios. The observed declines in debt-to-interest and debt-to-exchange rate ratios under pessimistic conditions were consistent with the modeled increases in the respective denominators, reinforcing the need for careful interpretation of such indicators. Overall, the scenario simulations validated the models' capacity to capture dynamic economic conditions, supporting their relevance as tools for fiscal planning and policy stress-testing in Kenya. The insights generated underscore the value of employing multiple modeling approaches in public debt forecasting. Machine learning tools can enhance accuracy in short-term projections, while classical econometric models contribute vital interpretability for policymaking as explained by Brockwell Davis, (2016); Varian, (2014).

Ultimately, Kenya's fiscal authorities can benefit from integrating both types of models in their planning frameworks. However, they should be cautious in interpreting results from black-box algorithms and always complement such outputs with economic theory and domain expertise.

5.2 Recommendations

Based on the findings from the forecasting models and analysis of macroeconomic determinants of public debt in Kenya, several important recommendations emerge for policymakers and future researchers. These suggestions aim to improve the accuracy and usefulness of public debt forecasting as well as to enhance fiscal policy effectiveness and the robustness of future academic work.

5.2.1 Policy Recommendations

It is advisable for Kenyan fiscal authorities to adopt a hybrid forecasting framework that combines machine learning techniques, such as XGBoost, with traditional econometric models like ARIMAX. This integrated approach capitalizes on the high predictive accuracy of machine learning while retaining the interpretability of time series models, thereby improving the precision and reliability of public debt projections. Embedding these models into fiscal planning processes will empower policymakers to anticipate debt trends more effectively and make informed decisions to maintain fiscal sustainability.

Given that exchange rate fluctuations significantly influence public debt levels, policymakers should focus on measures to stabilize the Kenyan shilling. This can be achieved through proper management of foreign exchange reserves, promoting export diversification, and reducing reliance on foreign currency-denominated debt. Stabilizing the exchange rate is crucial to mitigating risks related to currency depreciation, which can amplify debt burdens and expose the economy to external shocks. Inflation management remains a critical policy area, as rising inflation tends to increase nominal debt levels, therefore reducing fiscal space. Coordinated efforts between the government and the Central Bank to maintain inflation within target ranges will help preserve debt sustainability, enhance economic stability, and boost investor confidence. Interest rate policies also require careful calibration. While controlling interest rates can influence borrowing costs and debt accumulation, it is important to strike a balance to avoid hampering economic growth or increasing debt service obligations excessively. Prudent interest rate management will contribute to sound debt dynamics and overall economic health.

Finally, incorporating scenario-based fiscal planning into routine government processes is recommended. Scenario simulations allow policymakers to explore potential future economic conditions and stress test fiscal strategies against adverse developments. Institutionalizing such approaches will improve preparedness, risk assessment, and policy responsiveness.

5.2.2 Recommendations for Further Research

Future research should explore the application of alternative machine learning algorithms, including Long Short-Term Memory (LSTM) networks, Random Forests, and other hybrid models that may better capture non linearities and temporal patterns in macroeconomic data. These models could potentially enhance forecasting accuracy beyond what is achievable with current approaches.

Expanding the set of explanatory variables to include fiscal deficits, government revenue, political risk indicators, and external shocks would deepen the understanding of factors influencing public debt. Linking forecasting models more explicitly with debt sustainability frameworks and risk assessment tools represents another promising research direction. Integrating stress testing methods would help assess fiscal resilience under various adverse scenarios, enabling policymakers to devise more robust debt management strategies. Given the sometimes opaque nature of machine learning models, further work is needed to improve interpretability through techniques like SHAP values or LIME. Enhancing model transparency will increase trust and facilitate practical adoption in policy environments by clarifying the influence of different macroeconomic variables on debt forecasts.

Lastly, investigating how forecasting models perform during periods of structural breaks or economic crises would provide valuable insights into their robustness. Developing models capable of detecting and adapting to such breaks would improve forecasting reliability in times of volatility and uncertainty.

References

- Advanced Feature Engineering for Time Series Data <https://medium.com/40rahulholla1/advanced-feature-engineering-for-time-series-data-5f00e3a8ad29>
- Agung, I. G. N. (2009). Time Series Data Analysis Using EViews. John Wiley Sons.
- Ahmed, M. U., Rahman, M. G., Islam, M. T. (2020). A Hybrid Model for Time Series Forecasting Using ARIMA and XGBoost. International Journal of Advanced Computer Science and Applications, 11(6), 377–384. <https://doi.org/10.14569/IJACSA.2020.0110649>
- Badrawani, A. (2025). Forecasting Inflation Using XGBoost and SARIMA: A Case Study From Indonesia. Journal of Applied Economics and Finance, 12(1), 33–45.
- Barro R J (1979). On the Determination of the Public Debt. Journal of Political Economy, 87 (5, Part 1): 940-971 <https://doi.org/10.1086/260807>
- Bentéjac, C., Csörgő, A., Martínez-Muñoz, G. (2019). A Comparative Analysis of XGBoost. arXiv. <http://arxiv.org/abs/1911.01914>
- Box, G. E. P., Jenkins, G. M., Reinsel, G. C. (2015). Time Series Analysis: Forecasting and Control (5th ed.). Wiley.
- Brockwell, P. J., Davis, R. A. (2016). * Introduction to Time Series and Forecasting* (3rd ed.). Springer.
- Cheruiyot, B. K., Onyango, R. O., Mogesi, M. B., Kisina, M. M. (2024). Forecasting Inflation in Kenya using ARIMA(2,2,2). African Scientific Annual Review, 1(1), 1–15.
- Daniel, T. (2018). Forecasting U.S. Treasury Interest Rates using Machine Learning techniques: XGBoost and LSTM models. Journal of Financial Modeling and Analytics, 4(2), 44–57.
- Doshi-Velez, F., Kim, B. (2017). Toward a Rigorous Science of Interpretable Machine Learning. arXiv preprint arXiv:1702.08608. <https://doi.org/10.48550/arXiv.1702.08608>
- Dowoge Frankline Keraro, Zakay NiMans, Dominic M, and Martice Klanyonyi (2024) Forecasting Kenya's Public Debt using Time Series Analysis. International Journal of Advanced and Applied Sciences <https://www.escience-gate.com—IJAAS.html>
- Gujarati, D. N., Porter, D. C. (2009). Basic Econometrics (5th ed.). McGraw-Hill Education.
- <https://documents1.worldbank.org/curated/en/796991589998832687/pdf/Kenya-Joint-World->

Bank-IMF-Debt-Sustainability-Analysis.pdf

<https://kippra.or.ke/composition-of-domestic-debt-in-kenya-and-implication-on-refinancing-risk/>

<https://kippra.or.ke/exchange-rate-dynamics-and-its-implications-on-debt-and-trade-flows/>

<https://mujast.mtu.edu.ng/storage/issues/Year-2024-Vol-4/Number-2/1729800557-MUJAST-240801.pdf>

<https://www.sciencedirect.com/science/article/pii/S2666143823000042>

<https://www.businessperspectives.org/index.php/journals/public-and-municipal-finance/issue-324/government-debt-forecasting-based-on-the-arima-model>

<https://www.knbs.or.ke/wp-content/uploads/2024/05/2024-Economic-Survey.pdf>

<https://www.ml.cmu.edu/research/dap-papers/f18/dap-martin-daniel.pdf>

<https://worldbank.org/en/news/press-release/2021/06/08/world-bank-global-economic-prospects-2021>

Hyndman, R. J., Athanasopoulos, G. (2018). *Forecasting: Principles and Practice* (2nd ed.). OTexts

Institute of Economic Affairs (IEA Kenya) ieakenya.or.ke/blog/kenyas-public-debt-distress-issues-and-scenarios/

International Monetary Fund (IMF). (2023). **Fiscal Monitor: On the Path to Policy Normalization**. <https://www.imf.org/en/Publications/FM>

Jiang, C., Ma, X., Chen, Y. (2020). Exchange Rate Volatility and Government Debt Sustainability in Developing Countries. *Emerging Markets Finance and Trade*, 56(12), 2852–2868. <https://doi.org/10.1080/1540496X.2019.1700348>

Jongwanich, J. (2009). The Impact of Exchange Rate Volatility on the Real Sector A Comparative Analysis between Developed and Developing Countries. *Journal of International Development*, 21(2), 165–186.

Kashif, M., Ślepaczuk, R. (2024). LSTM-ARIMA Hybrid Models for Algorithmic Trading Strategy Enhancement: Empirical Evidence from Financial Markets. *Expert Systems with Applications*, 236, 121248. <https://doi.org/10.1016/j.eswa.2023.121248>

- Kavousi-Fard, A., Baziar, A., Abedi, M. (2018). A Hybrid Approach for Time-Series Forecasting Using ARIMA and XGBoost: Application in Energy Prediction. *Energy Reports*, 4, 944–955. <https://doi.org/10.1016/j.egy.2018.11.003>
- Kithure ME (2021). Use of Time Series Models in Forecasting of VAT in KRA. *African Tax and Customs Review*, 4 (2):22-22
- Lidiema, C., Waititu, A., Mageto, T., Ngunyi, A. (2018). Forecasting Household Credit in Kenya Using Bayesian Vector Autoregressive (BVAR) Model. *American Journal of Applied Mathematics and Statistics*, 6(1), 17–24.
- Maina, G. W., Ndiritu, J. N., Ouma, S. A. (2024). Dynamic Interaction between Public Debt and Inflation in Kenya: A VAR Approach. *African Journal of Economic Policy*, 31(1), 85–102.
- Masini, R. P., Medeiros, M. C., Mendes, E. F. (2020). Machine Learning Advances for Time Series Forecasting. *arXiv*. <http://arxiv.org/abs/2012.12802>
- Makau J, Ocharo K, and Njuru S (2018). Fiscal Policy and Public Debt in Kenya. *IOSR Journal of Economics and Finance*, 9(5) : 12 – 15
- Makridakis, S., Spiliotis, E., Assimakopoulos, V. (2018). Statistical and Machine Learning Forecasting Methods: Concerns and Ways Forward. *PLOS ONE*, 13(3), e0194889. <https://doi.org/10.1371/journal.pone.0194889>
- Mbiti, A., Njoroge, M. (2020). The Impact of Macroeconomic Variables on Public Debt Dynamics in Kenya. *International Journal of Economics and Finance*, 12(4), 112–121. <https://doi.org/10.5539/ijef.v12n4p112>
- Muhanji, S., Ojah, K. (2011). Management and Sustainability of External Debt: A Focus on the Emerging Economies of Africa. *Review of Development Finance*, 1(3–4), 184–198. <https://doi.org/10.1016/j.rdf.2011.09.002>
- Mukono, N. M. (2019). Forecasting the Term Structure of Government Bonds in Kenya (Master's thesis). University of Nairobi, Nairobi, Kenya.
- Mutuku, C. M., Koech, E. (2021). Forecasting Kenya's Public Debt using ARIMA and VAR Models. *Journal of African Macroeconomic Research*, 9(2), 58–75.
- Naseem, S., Liu, S., Ahmed, M. (2021). Financial Time Series Forecasting Using Machine Learning models. *Journal of Risk and Financial Management*, 14(7), 314. <https://doi.org/10.3390/jrfm14070314>

- Nichani, S., Patel, A., Singh, R. (2024). Optimizing Financial Time Series Prediction Using Hybrid ARIMA-LSTM-XGBoost Models. *Journal of Financial Data Science and Engineering*, 6(1), 21–34.
- Njenga, J. K. (2024). Domestic Credit Growth Analysis Using ARIMA Technique: A Case Study of Kenya. *Journal of Economics, Management and Trade*, 30(5), 12–21.
- OECD. (2020). *Economic Outlook No. 107*. Organisation for Economic Co-operation and Development. <https://www.oecd.org/economic-outlook/>
- Omondi, C. O., Waititu, A. G. (2024). Hybrid ARIMA-XGBoost model for Mobile Money Transaction Forecasting in Kenya. *African Journal of Statistics and Modeling*, 12(3), 45–58.
- Owaga, P. O. (2021). Factors Influencing Public Debt in Kenya (2000-2019). Unpublished master's thesis, KCA University, Nairobi, Kenya.
- R. Odegua (2020) Providing Bank Loan Default with XGBoost <https://www.researchgate.net/publication/339088951>
- Reinhart, C. M., Rogoff, K. S. (2010). Growth in A Time of Debt. *American Economic Review*, 100(2), 573–578. <https://doi.org/10.1257/aer.100.2.573>
- Shi, Y., Liu, Y., Zhou, Y. (2022). CNN-LSTM-XGBoost Model for Time Series Forecasting in Financial Markets: An Attention-Based Approach. *Applied Soft Computing*, 116, 108334. <https://doi.org/10.1016/j.asoc.2021.108334>
- Siami-Namini, S., Siami Namin, A. (2018). Forecasting Economics and Financial Time Series: ARIMA vs. LSTM. *arXiv*. <http://arxiv.org/abs/1803.06386>
- Statista (2024) [Statista.com/statistics/451122/national-debt-of-kenya-in-relation-to-greater-domestic-product-gdp](https://www.statista.com/statistics/451122/national-debt-of-kenya-in-relation-to-greater-domestic-product-gdp)
- Stepień, J., Ślepaczuk, R. (2025). Hybrid Machine Learning Models for Financial Forecasting: A Comparative Study on the SP 500 and Bitcoin. *Quantitative Finance Letters*, 4(1), 25–41.
- Tashman, L. J. (2000). Out-of-Sample tests of Forecasting Accuracy: An Analysis and Review. *International Journal of Forecasting*, 16(4), 437–450.
- Ugoh, S. C., Obike, A. E., Ujah, E. I. (2021). Forecasting GDP in Nigeria using ARIMAX model: A Study of Imports As an Exogenous Variable. *Nigerian Journal of Economic Modelling*, 9(1), 19–34.

- Varian, H. R. (2014). Big data: New Tricks for Econometrics. *Journal of Economic Perspectives*, 28*(2), 3–28. <https://doi.org/10.1257/jep.28.2.3>
- Verma P, Reddy SV, Ragha L, and Datta D (2021). Comparison of Time-series Forecasting Models. In the International Conference on Intelligent Technologies, IEEE, Hubli, India: 1- 7. <https://doi.org/10.1109/CONIT51480.2021.9498451>
- Wambugu, M. W. (2022). Forecasting Inflation in Kenya Using ARIMAX Models. *Kenya Journal of Economic Modeling and Policy*, 3(1), 33–45.
- World Bank, 2021; IMF, 2022 <https://worldbank.org/en/topic/debt>
- Zhang, G. P., Eddy Patuwo, B., Hu, M. Y. (2019). Forecasting with Artificial Neural Networks: The State of the Art. *International Journal of Forecasting*, 14(1), 35–62. [https://doi.org/10.1016/S0169-2070\(97\)00044-7](https://doi.org/10.1016/S0169-2070(97)00044-7)
- Zhou, L., Pan, W., Wang, S., Li, Y. (2021). Forecasting Government Debt with Machine Learning Techniques: Evidence From Emerging Economies. *Economic Modelling*, 99, 105493.
- Zhu, Y., Zhang, L., Fang, Z. (2023). A Hybrid ARIMA-XGBoost Model Based on Wavelet Decomposition for Commodity Price Forecasting. *Journal of Forecasting and Economics*, 18(4), 275–292.
- Zhuravka, F., Pavlenko, Y., Kravchuk, I. (2019). Modeling Ukraine's Government Debt Dynamics using ARIMA Models. *Financial and Credit Activity: Problems of Theory and Practice*, 4(31), 229–239. <https://doi.org/10.18371/fcaptp.v4i31.190664>
- Zulu, M., Mwansa, K. (2020). Time Series Modeling of Zambia's External Debt Using ARIMA, MA and GARCH Models. *Zambian Journal of Statistics and Policy*, 2(1), 73–89.